Master Thesis

in the study program User Experience Design
Faculty of Computer Science

Designing a Web-Based UAV Interface for Firefighting Dispatchers to Allow Information Retrieval During Multitasking

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Declaration

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Note on the Use of Artificial Intelligence

In this work, artificial intelligence tools were used to support the writing process. These tools were applied to correct grammatical errors, improve language clarity, and assist in identifying related academic literature.

All changes were made under the author's supervision. The content, structure, interpretation of results, and conclusions of this thesis remain the sole responsibility of the author. Artificial intelligence tools were not used to generate research ideas, design the study, perform data analysis, or draw scientific conclusions.

Abstract

Emergency dispatchers operate in multitasking environments in which incident information changes continuously and interruptions are frequent. Prior research on the use of Unmanned Aerial Vehicle (UAV)s in emergency response has largely focused on field operations, workload, and alerting systems, with limited attention to how dispatcher interfaces support interruption recovery and reconstruction of incident state. This thesis investigates how structured information retrieval support influences dispatcher interaction with UAV based incident information under interruption and multitasking conditions.

To address this problem, a dispatcher centered web based interface was designed to support re-engagement through chronological organization of events, filtering mechanisms, and summaries of missed updates. The interface was evaluated in comparison to a interface without information retrieval support in a controlled laboratory study using a within subjects design. A total of 29 participants completed simulated emergency monitoring tasks in both conditions. Measures included task performance, cognitive workload, situational awareness accuracy and response time, eye tracking data, and qualitative interview responses.

The results showed no differences between conditions in task completion time, subjective workload, or perceived usability. However, when structured information retrieval support was available, situational awareness accuracy during re-engagement was higher and response times to situational probes were lower. These findings indicate that structured access to incident information can support re-engagement after interruptions without increasing measured cognitive workload within the investigated experimental setting.

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1. Introduction

Emergency response systems operate under time pressure, uncertainty, and consequences of error [6, 64]. Dispatchers coordinate emergency operations and make decisions based on incomplete and changing information [70, 36]. A typical emergency response begins when a bystander reports an incident, such as a traffic accident on a motorway, to the emergency number 112. Based on this verbal report, the dispatcher must assess the situation, determine its severity, and decide which emergency units to deploy [10, 53]. At this stage, the dispatcher has no direct visual access to the scene and must rely on fragmented descriptions provided under stress [16, 55].

Dispatchers rarely focus on a single incident. They manage multiple incidents in parallel, answer new calls, coordinate responding units, and react to updates as they arrive [47, 66]. These frequent interruptions require dispatchers to repeatedly shift attention between incidents [57]. When returning to an ongoing incident, dispatchers must reconstruct Situational Awareness (SA) from memory and available system information [11, 70]. If relevant information is difficult to retrieve or poorly structured, this re-engagement process becomes slow and error prone [59].

Recent advances in sensing technologies, including the use of UAV, make it possible to provide dispatch centers with aerial imagery and scene level information [17]. Such data can support SA by offering perspectives that are otherwise unavailable from verbal reports alone [11].

This thesis focuses on information retrieval support for firefighting dispatchers working under multitasking conditions. It investigates how structured access to UAVs derived information can support SA and re-engagement after interruptions.

In this thesis, re-engagement refers to the interaction process that occurs when a dispatcher returns to an incident after an interruption and must regain an understanding of the incident state using the available interface information. The term is used to describe this interaction phase only and does not imply changes in decision authority or operational procedures.

1.1. Problem statement

Most previous research and operational UAV systems in emergency response are designed for field personnel or drone operators rather than dispatchers [2, 8, 23]. Dispatchers operate in a multitasking environment characterized by frequent interruptions and changing priorities, which differs from the context of UAVs pilots [36].

While UAVs can provide visual updates, existing interfaces primarily focus on live video feeds or static imagery and provide limited support for information retrieval [49]. When dispatchers return to an incident after an interruption, they often lack tools that help identify what has changed during their absence or filter information based on incident relevance. At the same time, additional information increases the amount of data that dispatchers must process [47]. Without mechanisms that support filtering, prioritization, and re-engagement, additional information can contribute to overload rather than improving decision making [57].

This results in two central challenges:

- Information overload: dispatchers may receive large amounts of UAVs data without contextual filtering.
- Situational gaps: after task switching, dispatchers struggle to reengage with ongoing incidents.

1.2. Research aim

The objective of this thesis is to design and evaluate a UAVs information retrieval interface for firefighting dispatchers working in emergency response centers. Unlike existing UAVs systems intended for drone pilots or field operators [17, 52], the proposed interface focuses on dispatchers who multitask and return to ongoing incidents after interruptions.

The interface is designed around three interaction features:

- Re-engagement support through summaries of changes that occurred during a dispatcher's absence.
- Prioritization of updates based on urgency.
- Filtering of information based on region, incident identifier, or time.

The study evaluates how these features influence dispatcher performance, reorientation after task switching, and usability under multitasking conditions

The theoretical foundations of SA, multitasking, interruption management, and information retrieval in safety critical systems are presented in Chapters 2 and 3. Based on these foundations, the formal research questions and hypotheses guiding the experimental evaluation are derived and introduced in Chapter 3.8.

1.3. Scope and contributions

This thesis contributes to research on dispatcher centered interaction with UAV based incident information in the following ways.

- It presents a dispatcher centered interface that supports information retrieval during multitasking and interruptions in emergency response contexts.
- It reports an empirical evaluation of how structured information retrieval influences dispatcher task performance, SA, cognitive workload, and perceived usability.
- It provides empirical evidence on re-engagement with incident information following interruptions in simulated UAV assisted dispatch scenarios.

The work focuses on firefighting dispatchers operating in emergency response centers and evaluates interaction with UAV derived information within a simulation environment. It does not address UAV autonomy, flight control algorithms, or interfaces for field responders. The findings are limited to the experimental conditions, tasks, and interface concepts investigated in this thesis.

1.4. Research approach

This thesis follows a controlled experimental approach to examine differences between interface variants under defined multitasking conditions. In addition, the interface design was informed by a user centred design approach to ensure that dispatcher tasks, information needs, and interruption related challenges were considered during development. The user centred design framework is used to structure the design process rather than as a formal evaluation method and is described in detail in the Design Process chapter.

The experimental approach focuses on interaction level outcomes and does not aim to evaluate operational performance or real world deployment. Details of the design process, experimental setup, tasks, and measures are provided in later sections of the thesis.

1.5. Research outline

This thesis is organized into the following chapters.

Chapter: 2 provides background necessary to contextualize the study. It introduces emergency dispatch operations, core concepts of SA, cognitive workload, and the role of UAV in safety critical systems. The chapter establishes conceptual foundations that inform the design and evaluation decisions made in later chapters.

Chapter: 3 reviews prior research relevant to this thesis. It covers work on emergency dispatching, UAV supported operations, interface design, and information management in multitasking environments. The chapter identifies limitations in existing approaches and positions the present work within the current research landscape.

Chapter: 4 describes the design process that led to the development of the dispatcher centered interface. It outlines how dispatcher context, task characteristics, and interruption related challenges informed design decisions. The chapter explains the derivation of interface requirements and the evolution of interface components used in the study.

Chapter: 5 presents the system architecture and implementation of the experimental platform. It describes the technical components used to support simulation, data logging, and interface interaction, and explains how these components enable controlled experimental evaluation.

Chapter: 6 details the research methodology and experimental design. It describes the study setup, participant tasks, experimental conditions, and measurement instruments used to assess performance, SA, workload, and usability. The chapter defines how data were collected and prepared for analysis.

Chapter: 7 reports the results of the experimental study. Quantitative and qualitative findings are presented without interpretation, following the predefined analysis plan. The chapter focuses on observed differences between interface conditions and measured outcomes.

Chapter: 8 discusses the results in relation to the research questions and prior work. It interprets the findings, reflects on their implications for interface design and dispatcher support, and considers limitations of the study.

Chapter: 9 concludes the thesis by summarizing the main contributions and findings. It outlines the limitations of the work and identifies directions for future research.

2. Background

This chapter introduces the operational, cognitive, and technological foundations required to understand the design and evaluation of a web based UAV interface for firefighting dispatchers. It first outlines the emergency dispatch context and dispatcher work practices, then reviews key concepts related to SA and cognitive workload under multitasking and interruptions. Finally, it discusses the use of UAVs in emergency response and relevant human system integration principles that inform dispatcher centered interface design.

This chapter does not aim to provide a comprehensive review of emergency management or decision theory. Instead, it introduces only those concepts that are directly relevant to understanding dispatcher interaction with information systems under interruption and multitasking conditions. The focus is on background knowledge required to motivate interface design decisions and evaluation measures used later in the thesis.

2.1. Operational context of emergency dispatch

This section describes the operational environment in which emergency dispatchers work. It outlines the organizational structure of dispatch centers and the cognitive demands associated with dispatcher workflows to establish the real world context for the interface design problem addressed in this thesis.

From an interaction perspective, dispatcher work involves continuous integration of information rather than sequential task completion. Dispatchers alternate between multiple ongoing incidents, each at a different stage of progression. This characteristic has implications for information access, since dispatchers frequently resume work on an incident after attending to other tasks.

2.1.1. Emergency response and dispatch centers

Emergency response relies on coordination between field teams and remote centers to manage tasks, resources, and information, but time pressure and uncertainty often make this coordination challenging [6]. A study of emergency medical services in Bavaria found that 28. 9% of ambulance deployments ended without patient transport, indicating that dispatchers often make decisions without complete information from the scene. Regional differences in transport rates reached up to 16. 8% points for certain incident types, such as fire alarm system calls, highlighting how dispatch decisions depend heavily on the availability of accurate SA and structured assessment during the emergency call [10].

In Germany, emergency dispatch is handled through Integrated Control Center (ILS), which coordinate Emergency Medical Services (EMS), firefighting units, and other responders [34] . These centers rely on Computer Aided Dispatch (CAD) systems to support decision making and resource allocation under time pressure [21] .

2.1.2. Dispatcher workflows and cognitive demands

Dispatchers operate in safety critical environments that require continuous monitoring, rapid decision making, and coordination across multiple information sources [47]. They interpret incoming information, manage competing priorities, and respond to evolving situations under uncertainty. These demands place high cognitive requirements on dispatchers, particularly when information is incomplete or fragmented.

The effectiveness of dispatch operations depends not only on data availability but also on how information is structured and presented. As noted by Wong and Blandford [70], decision making in such environments relies on interfaces that support rapid interpretation and action rather than raw data access alone.

Information relevant to an incident may arrive asynchronously and vary in relevance over time. Dispatchers therefore reassess previously received information and integrate new updates into an existing mental representation of the incident. These demands increase when dispatchers return to an incident after an interruption and reestablish SA based on available system information.

2.2. Cognitive foundations of dispatcher work

Effective interface design for emergency dispatch requires an understanding of the cognitive processes involved in decision making under time pressure. This section reviews key theoretical concepts related to SA, cognitive workload, multitasking, and interruptions, which are central to understanding dispatcher performance and re-engagement challenges.

2.2.1. Situational awareness in safety critical systems

SA is a central concept in safety critical systems such as emergency dispatch centers. Endsley defines SA as a three level process consisting of the perception of relevant elements in the environment, the comprehension of their meaning, and the projection of their future status [11]. For dispatchers, this involves perceiving incoming information from callers and sensors, understanding how this information relates to the incident, and anticipating how the situation may evolve.

In emergency dispatch contexts, the three levels of SA are rarely supported within a single interface view. Perception may be supported through incoming information streams, while comprehension and projection require integration across multiple sources and time points. Interruptions can disrupt this integration process, requiring dispatchers to reconstruct SA when returning to an incident.

SA depends strongly on system design. Poorly designed systems that lead to fragmented or delayed information can reduce SA and negatively affect decision making performance [12]. In emergency dispatching, reduced SA can lead to delayed responses, misallocation of resources, or inappropriate prioritization.

Figure 1 illustrates the three level model of SA as described by Endsley.

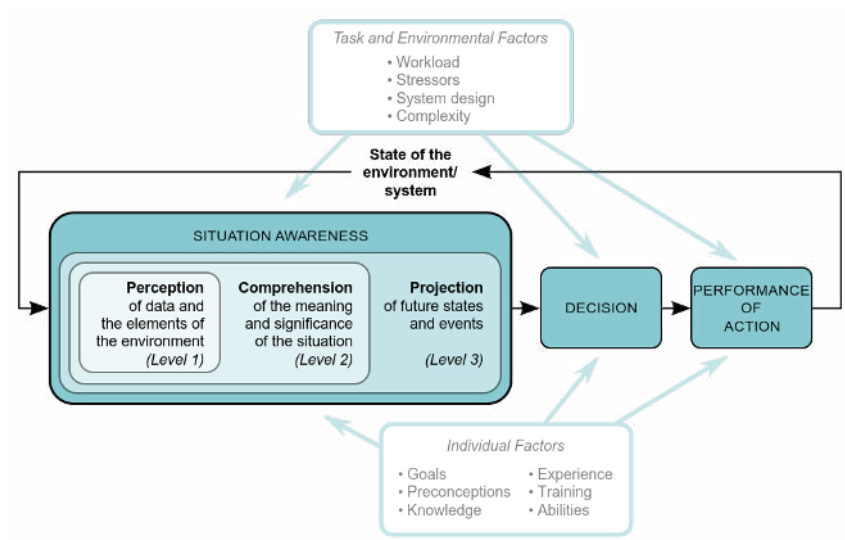

Figure 1: Conceptual model of SA showing perception, comprehension, and projection, and their relationship to decision making and performance, adapted from Endsley [11]. Image source: Wikimedia Commons.

2.2.2. Cognitive workload, multitasking, and interruptions

Cognitive workload is a central factor in dispatcher performance under multitasking and interruption conditions [30] . This section reviews research on workload, task switching, and interruptions in safety critical systems and discusses their implications for dispatcher interaction with incident information.

Safety critical systems require effective human machine interaction to manage time critical decisions, information overload, and safety risks [37]. Dispatchers frequently work under high cognitive workload due to multitasking requirements, frequent interruptions, and the need to switch attention between concurrent incidents.

Interruptions and task switching can disrupt cognitive processes and make it difficult for dispatchers to maintain or regain SA[72]. When information is not structured to support re-engagement, these disruptions can increase cognitive load and negatively affect performance.

Beyond increased attentional demand, interruptions introduce a resumption phase in which users reestablish task context before continuing. During an interruption,task relevant information may decay in working memory, and returning users reconstruct the incident situation by combining memory with external system cues [32]. In dispatcher work, this reconstruction is required repeatedly because task switchingis not occasional but routine.

The cognitive cost of interruptions is therefore not limited to the interruption itself, but also includes the effort required to identify what has changed during the period of absence [32]. When updates accumulate while attention is directed elsewhere, operators compare past and current system states, locate missed information, and judge relevance [32, 13]. If interfaces do not represent changes explicitly, re-engagement depends more heavily on memory-based reconstruction and visual search processes, which can increase cognitive workload even when the underlying task remains unchanged [57].

From a design perspective, these mechanisms imply that interruption recovery should be treated as a distinct interaction phase. Interfaces that provide structured access to incident history and highlight missed updates can reduce the need for manual search and support faster rebuilding of SA after attention shifts [32, 1].

In dispatcher work, workload is influenced not only by the amount of available information, but also by the effort required to locate, interpret, and relate information across tasks. When dispatchers manually search for missed updates or reconstruct incident state after interruptions, cognitive workload can increase even if task demands remain constant [51, 35]. This highlights the importance of interface support that reduces search effort and externalizes information needed for re-engagement.

2.3. Technological and design context

Advances in UAV technology provide new opportunities to support SA during emergency response. This section reviews existing uses of UAVs in emergency contexts and discusses human system integration and interface design principles relevant to presenting UAV derived information to dispatchers.

For dispatchers, the primary challenge is not access to UAV data but integration of that data into existing multitasking workflows. Interfaces that assume continuous monitoring may conflict with dispatcher responsibilities across concurrent incidents. This highlights the need for designs that support selective access and re-engagement rather than sustained attention.

2.3.1. UAVs in emergency response

The integration of UAVs into emergency and disaster response has been shown to support SA by providing real time aerial imagery and sensor data from locations that may be difficult or unsafe to access directly [39]. UAV based mapping systems can process data locally at disaster sites, enabling responders to access three dimensional visualizations and object detection results without relying on remote infrastructure [23].

In Germany, initiatives such as the German Rescue Robotics Center have demonstrated the use of UAVs for rapid scene assessment, logistical planning, and coordination with ground units [62, 38]. To enable early UAV deployment during incident mobilization, approaches such as drone ports near incident locations have been proposed to reduce response latency and support timely availability of aerial data [5].

In these settings, UAVs typically perform flight related tasks autonomously, while human operators supervise and make decisions. This aligns with supervisory control models, where autonomous systems execute tasks independently and humans intervene when required [71].

2.3.2. Human system integration and interface design principles

Human System Integration (HSI) refers to the systematic consideration of human capabilities, limitations, and task demands in system design to support safe and effective performance in complex environments [37]. In safety critical domains, interface design must align with cognitive processes and operational workflows to support decision making under time pressure.

Research in human computer interaction emphasizes the role of external representations in supporting cognition [24]. Externalizing task relevant information can reduce reliance on working memory and support distributed cognitive processes across users and artifacts. Persistent representations of system state and event history allow users to reconstruct context when attention has shifted.

Studies on interruptions and task resumption show that returning users must retrieve prior task context and identify changes that occurred during absence [32, 57]. When changes are not explicitly represented, users rely more heavily on memory and visual search. Interfaces that represent event history, highlight updates, or structure information chronologically can support task resumption and reduce resumption time [32, 1].

Research on situational awareness further indicates that interface design influences perception, comprehension, and projection processes [11, 12]. Fragmented or poorly structured information can impair the integration of system elements required to maintain situational awareness.

In the context of UAV supported emergency response, these findings suggest that dispatcher interfaces should support integration of aerial information into existing coordination workflows. Since dispatchers alternate between incidents and experience frequent attention shifts, interface design must support information retrieval and reconstruction of incident state across time.

3. Related Work

This chapter reviews prior work relevant to the use of UAV systems in emergency response and dispatch contexts. The review progresses from UAV applications in field operations to control center integration, dispatcher decision making, and information management approaches. The goal is to identify how existing solutions address information access and where limitations remain for dispatcher multitasking and re-engagement.

3.1. UAV applications in disaster and firefighting scenarios

Research on UAV deployment in disaster and firefighting contexts has primarily focused on supporting responders in the field through enhanced situational information and remote sensing capabilities. Across this body of work, three recurring objectives can be identified: (1) spatial mapping and damage assessment, (2) real-time aerial surveillance, and (3) coordination support for on-site tactical decision making.

Boccardo et al. [2] demonstrated that UAVs can generate accurate geospatial mappings of emergency sites, supporting responders in assessing structural damage and terrain conditions. Similarly, Jin et al. [29] examined how drone deployment strategies vary depending on disaster type and environmental constraints, emphasizing optimization of flight paths and sensor use. Stampa et al. [58] proposed a multi UAV monitoring system for large-scale industrial incidents, focusing on aerial data acquisition and coordinated surveillance.

While these studies demonstrate the operational value of UAVs in emergency response, they share a common interaction assumption: the primary users are field responders or mission operators who directly monitor drone outputs and engage in continuous visual supervision [23]. Interaction models in these systems are typically designed around sustained attention to live video streams, spatial navigation interfaces, and real time coordination with on site teams.

This user model differs fundamentally from the working conditions of emergency dispatchers. Dispatchers are not physically present at the incident site and do not supervise flight trajectories or sensor configurations. Instead, they manage multiple concurrent incidents under frequent interruptions and time pressure [30]. Their interaction with incident information is episodic rather than continuous and is oriented toward assessment, prioritization, and coordination across units rather than tactical on site action [70].

Consequently, although existing research demonstrates how UAVs enhance situational information for field operations, it provides limited insight into how drone derived information should be structured, filtered, and presented within interruption prone dispatch environments. The focus on real time surveillance and deployment optimization leaves open the question of how UAV data can be integrated into dispatcher centered workflows that require selective access, rapid re-engagement, and structured information retrieval.

3.2. UAV data use in this centers and dispatching

Kristiansson et al. [31] explored the integration of UAV data into dispatch workflows by providing drone images directly to dispatch centers before emergency services arrived. Although this approach enabled earlier access to visual information, the images were static and lacked interactivity or filtering mechanisms, limiting their usefulness during active dispatch operations. Ramchurn et al. [43] proposed a UAV human collaboration system focused on task delegation and coordination but did not address dispatcher oriented interaction with UAV data for decision support.

These approaches represent early steps toward integrating UAV data into dispatch workflows. However, they provide limited support for information retrieval during multitasking, particularly when dispatchers return to an incident after an interruption. This limitation is especially relevant in dispatch settings, where interaction with incident information is episodic rather than continuous.

These approaches represent early steps toward integrating UAV data into dispatch workflows. Recent work in German control centers has examined how automated UAV deployment can be embedded into ILS to support dispatch decision making during the early phases of emergency response [5]. Similarly, real world deployments have demonstrated that drones can transmit on site imagery directly to dispatch centres prior to the arrival of emergency medical services, thereby providing earlier situational input [31]. However, while these projects validate technical feasibility, they primarily focus on deployment and transmission mechanisms rather than structured information retrieval during multitasking. As noted in discussions on drone integration into emergency frameworks, effective incorporation requires substantial operational adaptation and workflow redesign beyond the mere addition of aerial data streams [54]. This limitation is particularly relevant in dispatch environments, where interaction with incident information is episodic rather than continuous.

3.3. Dispatcher decision errors and information overload

Dispatcher decision making involves managing multiple tasks under time pressure and uncertainty, often relying on experience, intuition, and simplified mental models [27]. Prior work has emphasized the importance of clear visual cues and information filtering to reduce cognitive load during emergency operations [40, 8].

Waal Nilsbakken et al. [65] investigated dispatch prioritization for trauma cases in Norway using a dataset of 3633 cases. The study reported significant dispatch errors, with 15 percent of serious injuries receiving low priority and 89 percent of minor cases receiving unnecessarily high priority. Elderly and female patients were disproportionately affected. These findings highlight how information overload and inadequate prioritization mechanisms can lead to systematic decision errors in dispatching contexts.

Importantly, these findings do not indicate individual failure but reflect structural challenges in managing information under time pressure. They suggest that dispatcher errors are influenced by how information is presented and prioritized rather than by a lack of available data.

Prior research further indicates that dispatcher decision making is shaped by how information is structured and accessed over time. In interruption prone environments, maintaining a coherent understanding of incident state becomes difficult when information is fragmented across multiple sources or presented without temporal [13] . Studies in emergency management and human factors indicate that frequent task switching reduces the ability to integrate new information with previously observed cues, increasing reliance on partial signals and heuristics [40, 8].

Taken together, these findings highlight that information overload in dispatch settings is not solely a function of information volume. Instead, it reflects challenges in identifying what information is new, relevant, and actionable after interruptions. This perspective emphasizes the need for interface support that makes changes in incident state explicit, rather than assuming continuous monitoring or relying on memory based reconstruction.

3.4. Ground control stations and UAV interfaces

Commercial ground control stations such as QGroundControl and QBase 3D are designed to support operators who directly supervise and control UAV missions [44, 18]. These systems provide mission planning tools, telemetry dashboards, real time video feeds, and configuration interfaces for managing flight parameters, payload settings, and communication links. Their primary objective is to enable supervisory control of one or multiple UAVs during mission execution.

Standard GCS architectures are structured around a continuous supervisory control model. Interaction is organized at a vehicle level, where each UAV is treated as an individual controllable entity with associated flight parameters, trajectories, and mission configurations. Operators define mission objectives, assign tasks, configure vehicle properties, select flight profiles, and monitor execution through persistent telemetry streams and waypoint based flight maps. Even extended frameworks that integrate automated mission planning and decision support systems assume that the operator evaluates and approves vehicle level plans [44].

As shown in Figure 2, interaction is centered on mission authoring and UAV configuration. The interface emphasizes task dependencies, vehicle assignments, and spatial planning tools. Information is structured around aircraft supervision rather than around incident level summaries or episodic information retrieval.

From a cognitive perspective, these systems assume sustained attention to a mission and direct responsibility for flight level decisions. The operator monitors aircraft parameters, supervises trajectory execution, and intervenes when deviations occur. Interaction is continuous and control oriented.

Figure 2: Mission designer interface in QGroundControl illustrating map based mission modeling, task configuration, vehicle assignment, and dependency management.
Source: QGroundControl Documentation, https://www.mdpi.com/1424-8220/18/7/2339

In contrast, emergency dispatchers operate under frequent interruptions and manage multiple concurrent incidents [36, 57]. They do not configure flight profiles or supervise trajectory execution. Instead, UAV derived information serves as an additional information source that supports incident assessment and coordination decisions. Interaction with UAV data is intermittent and interruption driven rather than continuous and control focused.

This difference in task structure creates a structural misalignment between conventional ground control station interfaces and dispatcher requirements. While GCS tools provide detailed flight telemetry and mission configuration capabilities, they offer limited support for structured incident level information retrieval, filtering by incident context, or summarizing changes that occurred during periods of inattention. Consequently, although existing systems are well suited for UAV pilots and mission supervisors, they do not directly address the cognitive and workflow demands of emergency dispatchers managing evolving incidents. This gap motivates the development of a dispatcher centered interface focused on structured information retrieval and re-engagement support.

3.5. Filtering, prioritization, and information management approaches

Filtering mechanisms have been shown to improve dispatcher performance during high stress and high volume periods. Penverne et al. [41] introduced a two level call filtering system at an emergency communication center in France during the Coronavirus Disease 2019 (COVID19) pandemic. The system enabled first line operators to rapidly identify life threatening emergencies while forwarding less urgent calls to secondary handlers. This approach significantly improved call answer times despite increased call volume, demonstrating the effectiveness of structured filtering for managing information overload. Beyond call handling, information management research emphasizes the role of structured presentation in supporting operator performance under

cognitive load. Techniques such as temporal grouping, hierarchical organization, and selective disclosure have been shown to reduce attentional demand by allowing users to focus on subsets of information while preserving access to the broader context [42]. In high workload environments, these mechanisms help operators manage fluctuating demands without requiring constant visual scanning.

However, most existing filtering and prioritization approaches are designed for communication flow, case triage, or resource allocation rather than for interaction with continuously evolving sensor data. As a result, there is limited guidance on how filtering and information retrieval mechanisms should be applied to dispatcher facing UAV interfaces that present dynamic video and event information during multitasking.

3.6. Limitations of existing solutions for dispatcher multitasking

Across existing research and operational systems, UAV interfaces and control tools are rarely designed with dispatcher multitasking in mind. Most solutions assume continuous attention to UAV data or are tailored for drone pilots rather than control center operators. Interfaces specifically designed to support dispatchers in managing UAV information alongside parallel tasks and re-engagement after interruptions remain unexplored.

3.7. Summary and refined research gap

This review shows that while UAV technologies provide valuable capabilities for emergency response, existing systems do not adequately support the needs of emergency dispatchers. Much of the existing work focuses on field operations, direct UAV piloting, or high level coordination, with limited attention to dispatcher interaction under multitasking and interruption conditions.

Dispatchers frequently manage parallel incidents, respond to interruptions, and re engage with evolving situations based on incomplete or outdated information. Current UAV interfaces and control tools typically assume continuous attention to sensor data or are designed for users whose primary task is UAV operation. Filtering and prioritization mechanisms are often applied at the level of call intake or task assignment rather than at the level of ongoing information access.

Taken together, these limitations indicate a gap in existing research on how dispatcher facing UAV interfaces can support structured information access and re-engagement during multitasking. Addressing this gap requires systematic investigation of interaction mechanisms that support structured information access and re-engagement after interruptions in dispatcher multitasking environments. To examine these mechanisms empirically, the following research questions and hypotheses are formulated.

3.8. Research questions and hypotheses

The research questions were derived from established principles in human computer interaction and emergency response research. Prior work on dispatcher decision making, multitasking, and interruptions identifies recurring challenges related to SA , information retrieval, and cognitive workload in time critical environments [11, 57, 36]. In line with user centered design principles, the formulation of the research questions was guided by an analysis of dispatcher tasks,

information needs, and re-engagement demands rather than by system capabilities alone [28, 12].

In addition, general research design guidance emphasizes that research questions should be feasible, well scoped, and grounded in the practical problem under investigation rather than formulated in abstraction [14]. Accordingly, the questions focus on measurable effects of interface support on **task performance**, re-engagement, **cognitive workload**, and **usability**, which are commonly evaluated outcomes in human computer interaction studies of safety critical systems [67, 40].

Dispatcher performance

Research Question 1: Does a UAV information retrieval interface (WIR) affect dispatcher **task performance** compared to an interface without information retrieval support (WOIR)?

H1: Dispatcher **task performance** is higher in the With Information Retrieval (WIR) interface than in the Without Information Retrieval (WOIR) interface.

Re-engagement with incident information.

Research Question 2: Can an information retrieval interface (WIR) support dispatcher re-engagement with incident information after interruptions or task switching, as reflected by **situational awareness accuracy** and **response time**, compared to an interface without information retrieval support (WOIR)?

- **H2a:** Dispatcher **situational awareness accuracy** during re-engagement is higher in the WIR interface than in the WOIR interface.
- **H2b:** Dispatcher re-engagement **response time** is lower in the WIR interface than in the WOIR interface.

Workload and usability

Research Question 3: How does the information retrieval interface (WIR) affect dispatcher **cognitive workload** and perceived **usability** during multitasking scenarios, compared to an interface without information retrieval support (WOIR)?

- **H3a:** Dispatcher **cognitive workload** is lower in the WIR interface than in the WOIR interface.
- **H3b:** Perceived **usability** is higher in the WIR interface than in the WOIR interface.

4. Design Process

This chapter describes how the UAV dispatcher interface was developed from an understanding of dispatcher work to the interface variants evaluated in the study. The design process focused on supporting information retrieval and re-engagement after interruptions in dispatcher work.

The aim of the design was not to replicate or improve an operational control room system, but to construct an interface for experimental investigation of re-engagement behaviour. Design decisions were driven by dispatcher context, task characteristics, and experimental requirements.

4.1. User centred design framework

The design process followed a user centred design framework as defined in ISO 9241 210 [28]. This framework structures design as an iterative process that includes understanding the context of use, specifying user requirements, producing design solutions, and refining these solutions through iteration.

In this thesis, the framework was used to structure the design process rather than as a formal evaluation cycle. Analysis of dispatcher work corresponds to understanding the context of use. Scenario based task focus and derived requirements correspond to specification of user requirements. Development and refinement of interface components correspond to production of design solutions.

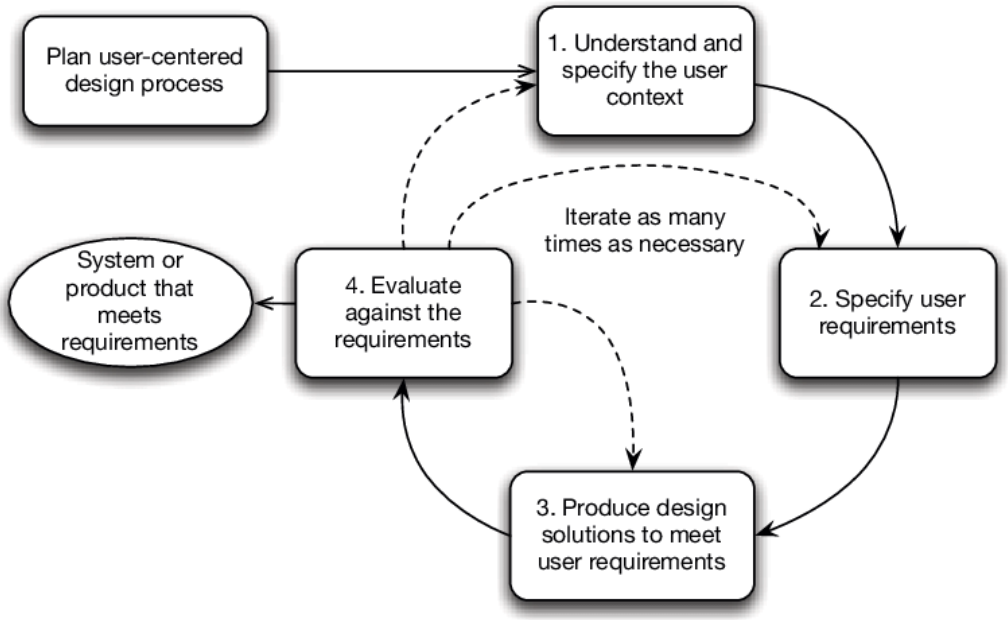

Figure 3: User centred design process according to ISO 9241 210 used to structure the design process in this thesis.

4.2. Understanding the context of use in integrated control centers

Emergency dispatchers in ILS operate under time pressure, parallel task handling, and frequent interruptions [5]. Within the ILS, dispatchers function as central coordination nodes. They receive emergency calls, assign operational keywords, dispatch units from fire and EMS, and continuously update the digital situation map through the CAD system.

Empirical observations of dispatcher work show that multiple communication channels and system interfaces be managed simultaneously [5]. Information from callers, field units, mapping systems, and documentation tools be integrated in real time. During early operational phases,

visual confirmation from the scene is typically unavailable, and updates are entered manually based on verbal reports. This contributes to information uncertainty and frequent shifts of attention between incidents and system components.

The integration analysis further indicates that dispatchers often manage several incidents concurrently while maintaining communication with field commanders and updating shared representations of incident state. These structural characteristics make interruptions inherent to dispatcher work rather than exceptional events. Consequently, interface concepts that assume continuous monitoring of UAV data are misaligned with operational reality.

When attention shifts away from an incident, dispatchers later reconstruct the current state of that incident based on accumulated updates and partial cues. Reestablishing SA under these conditions represents a recurring cognitive demand within ILS operations. This analysis defines the context of use as required in the user-centred design process according to ISO 9241-210.

4.3. Specification of user requirements

Following the identification of the context of use in Section 4.2, the next phase of the user centred design process according to ISO 9241-210 is the specification of user requirements. To derive these requirements, dispatcher interaction was abstracted into a recurring task pattern that reflects structural characteristics of integrated control center work.

The user requirements defined in this section were derived analytically from the identified context of use and from this task abstraction, rather than from empirical requirement elicitation methods such as interviews or participatory design activities.

4.3.1. Task abstraction: Re-engagement after interruption

Building on the dispatcher context described previously, the present work abstracts a recurring interaction unit: returning to an evolving incident after an interruption. Rather than modelling complete operational workflows, the design analysis concentrates on moments in which attention shifts away from an incident and later returns.

In integrated control centers, interruptions arise from incoming calls, coordination activities, parallel incident management, and monitoring responsibilities [5]. During periods of absence, incident information may continue to evolve through new reports, sensor detections, or updates from field units. Upon returning, dispatchers determine what has changed, what remains relevant, and how new information relates to their prior understanding of the situation.

This reconstruction process imposes cognitive demands that are not present during uninterrupted monitoring. Reestablishing SA requires structured access to accumulated updates and support for identifying changes in incident state. By centering the design analysis on these re-engagement moments, the present work isolates a core interaction challenge that emerges directly from dispatcher multitasking and interruption patterns.

In the context of UAV integration, this interaction pattern becomes particularly critical. Unlike traditional verbal reports, UAV systems generate continuous visual and event based data streams. Because dispatchers cannot maintain uninterrupted monitoring while managing parallel incidents, relevant updates may accumulate during periods of absence. Without structured support, re-engagement requires manual scanning of video feeds, event logs, or map layers to

identify what has changed. This increases cognitive load and risks overlooking time critical developments.

The abstraction of re-engagement as a recurring task unit therefore provides the analytical basis for deriving concrete user requirements for dispatcher facing UAV interfaces.

4.3.2. Derived design requirements

Based on the identified context of use and the task abstraction described above, concrete user requirements were derived in accordance with the user centred design framework.

The analysis demonstrates that interruption and parallel task handling are structural characteristics of dispatcher work rather than exceptional conditions. When integrating UAV systems that generate continuous visual and event based data streams, the interface support controlled access to accumulated information and facilitate rapid reestablishment of SA.

From this analysis, three design requirements were defined:

- **R1 Support re-engagement after interruptions.** The interface shall enable dispatchers to identify updates that occurred during periods of absence and reduce the need for manual reconstruction of incident state.
- **R2 Support filtering of UAV derived information.** UAV data shall be structured and selectively accessible to allow dispatchers to focus on relevant subsets of information while managing parallel incidents.
- **R3 Support visual differentiation of events.** Incident updates shall be visually distinguishable to support rapid orientation and reduce visual scanning effort, without embedding semantic prioritization or automated decision logic.

These requirements address interaction level challenges emerging from structural interruption patterns in ILS operations. They define the scope of the interface and serve as the basis for the design solutions described in the following chapter.

4.4. Design scope and boundary conditions

The design process was conducted within clearly defined scope boundaries to ensure alignment with the experimental goals of this thesis. The interface was conceived as a research instrument intended to support controlled investigation of re-engagement behaviour rather than as an operational decision support system.

Several design directions were explicitly excluded. The interface does not implement automated prioritization, semantic weighting, or decision recommendations. No interpretation of incident severity, urgency, or correctness is performed by the system. All presented information remains descriptive and observational, leaving assessment and decision making entirely to the participant.

This boundary was necessary to avoid introducing confounding variables that could obscure the effects of information retrieval support. Automated prioritization or recommendation logic would have altered task structure and cognitive demands in ways that are difficult to control experimentally. By restricting the design to visual structuring, filtering, and temporal organization of

information, the interface isolates re-engagement support without embedding assumptions about dispatcher decision strategies.

The design also avoids modelling organizational procedures, communication protocols, or collaboration between dispatchers. These aspects are central to real dispatch work but were outside the scope of the present study. Limiting the design to individual interaction with incident information supports experimental clarity while preserving relevance to dispatcher centred interaction challenges.

4.5. Design evolution and iterations

The requirements were explored through iterative development of interface concepts that support information access and re-engagement. Multiple interface variants were developed and refined to address the identified requirements.

Early concepts examined different ways of presenting UAV derived information in a form that remains interpretable after interruptions. Through iteration, three interface components emerged as relevant for re-engagement support. These components were a temporal representation of events, an event feed, and a summary view.

The components were refined to manage information presentation, distinguish past and new information, and support orientation after returning to an incident. Refinement was guided by alignment with experimental goals and supervisory feedback to ensure that each component contributed to the intended experimental manipulation.

Iteration was constrained by the need to maintain a stable task structure across experimental conditions. Changes were evaluated primarily in terms of whether they supported clearer access to incident updates after interruptions without introducing additional interaction steps or altering task goals.

As a result, iterations focused on representation and visibility rather than functionality. Components were refined to improve temporal clarity, visual distinction, and information grouping, while interaction mechanisms remained deliberately simple. This approach ensured that differences observed during evaluation could be attributed to information retrieval support rather than to changes in task execution or interface complexity.

Design refinement therefore emphasised consistency and comparability over feature expansion. This constraint reflects the role of the interface as an experimental stimulus rather than as a production system and guided the final selection of interface components used in the study.

The derived design requirements guided the selection of interface components. The timeline and event feed address R1 by providing access to information accumulated during dispatcher absence. Filtering mechanisms in the event feed address R2 by allowing selective inspection of UAV derived information. Visual differentiation across all components addresses R3 by supporting orientation without embedding semantic interpretation.

4.6. Interface design

The final design consists of two interface conditions that were created to support experimental investigation of information retrieval and re-engagement. Both conditions are based on the same overall layout, visual design, and interaction structure to ensure comparability. The conditions

differ only in the availability of components that support structured access to UAV derived information after interruptions.

In the WIR condition, the interface includes a timeline together with an event feed and a re-engagement summary, enabling dispatchers to access updates that occurred during their absence. In the WOIR condition, the interface provides access only to the timeline, requiring dispatchers to manually inspect past events to reconstruct incident state after an interruption.

Maintaining equivalence between interface conditions was a central design requirement. Both variants were designed to be visually and structurally identical except for the presence or absence of information retrieval components. This equivalence ensures that observed differences during evaluation cannot be attributed to layout changes, interaction mechanics, or visual prominence unrelated to re-engagement support.

By holding layout, visual encoding, and navigation constant, the design isolates the availability of structured information as the sole experimental manipulation. This supports interpretation of results in later chapters without introducing ambiguity regarding the source of observed effects.

4.6.1. Interface variants

To support experimental investigation of information retrieval and re-engagement, two interface variants were derived from the same base design. Both variants share identical layout, visual styling, and interaction structure to ensure comparability. They differ only in the availability of information retrieval components.

As shown in Figure 4, the WIR interface includes the event feed and re-engagement summary in addition to the timeline. These components provide structured access to events that occurred during dispatcher absence and reduce the need for manual reconstruction of incident state.

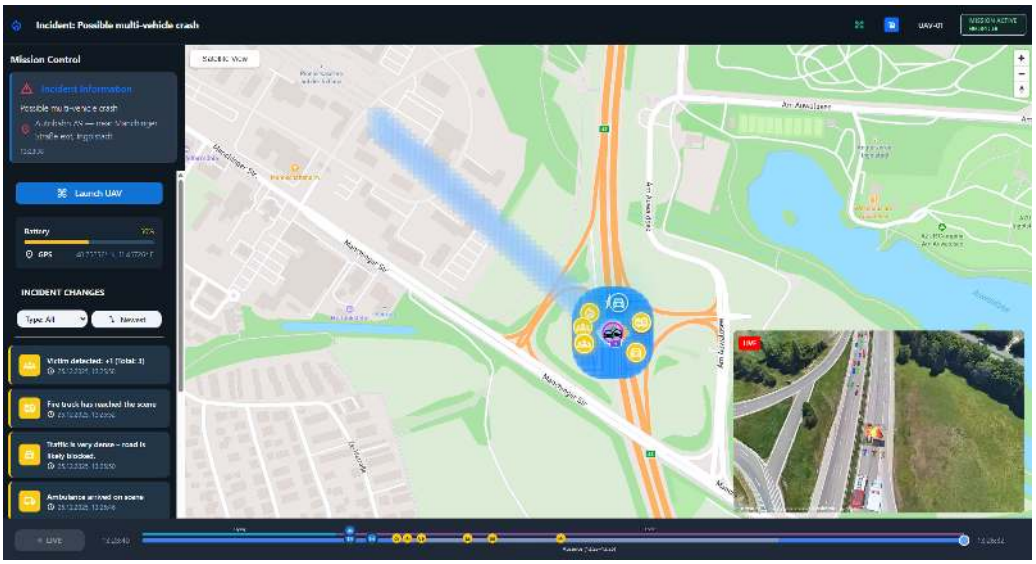

Figure 4: WIR interface showing the timeline together with the event feed and re-engagement summary.

As shown in Figure 5, the WOIR interface excludes the event feed and re-engagement summary. In this variant, dispatchers have access only to the timeline and manually inspect past events to determine what occurred during an interruption.

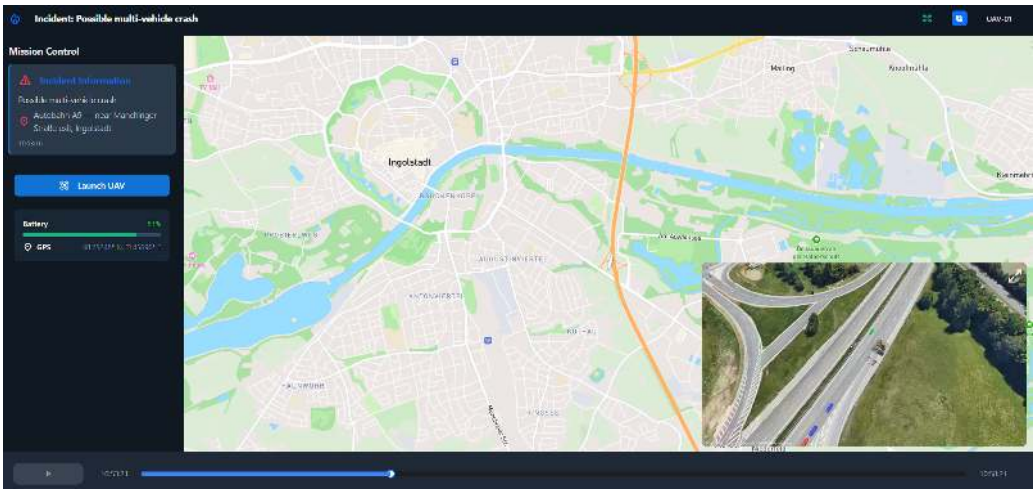

Figure 5: WOIR interface showing the timeline without additional information retrieval components.

4.6.2. Event feed

The event feed presents UAV derived information as a structured list of events associated with the incident. It supports inspection and filtering of information during and after interaction with the interface.

Events are displayed in a list ordered by time of occurrence and grouped by event type. The event feed provides a list based representation of the same event data shown in the timeline and visually distinguishes between events observed during interaction and events that occurred during an interruption, as shown in Figure 6. Events observed during interaction are shown in blue, while events that occurred during an absence are shown in yellow.

The list based representation was chosen to support systematic inspection of incident updates following interruptions. While the timeline provides an overview of temporal progression, the event feed allows dispatchers to review discrete events in a compact and stable format that supports comparison and selective inspection without requiring continuous temporal navigation.

Filtering controls allow dispatchers to adjust the event feed according to their current information needs. Events can be ordered chronologically to support inspection of recent or earlier updates, and event types can be selectively shown or hidden to focus on specific categories such as person detections. These controls support manual exploration of incident information while preserving access to the complete event history. The filtering mechanism affects only visibility and ordering and does not perform interpretation, prioritization, or ranking of events.

Visual differentiation in the event feed is used to support orientation between events observed during interaction and events that occurred during dispatcher absence. Color is applied solely to distinguish these states and does not convey semantic meaning regarding importance, urgency, or correctness.

The event feed complements the timeline by providing an alternative representation of the same underlying event data. Both components were included to support different information access strategies while maintaining identical data availability across interface conditions.

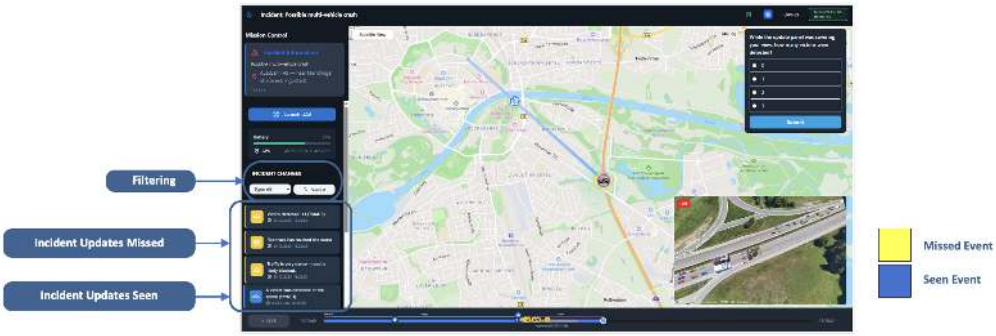

Figure 6: Event feed component showing filtering controls and visual distinction between events observed during interaction and events occurring during interruptions.

4.6.3. Timeline

The timeline presents UAV derived events in temporal order to support re-engagement after interruptions. It provides an overview of incident progression without requiring continuous monitoring.

Events are displayed chronologically, allowing distinction between information observed during interaction and information that occurred during absence, as shown in Figure 7. The timeline supports inspection of recent updates while retaining access to earlier events.

The temporal representation was chosen to support reconstruction of incident progression following interruptions. When returning to an incident, dispatchers understand not only which events occurred, but also their temporal order and relation to prior information. The timeline supports this reconstruction by providing a continuous temporal context that situates missed updates within the broader incident history.

The timeline was designed as a passive overview component that does not demand interaction or attention during task execution. Unlike alert based or notification driven approaches, the timeline does not interrupt ongoing activity. This design choice preserves dispatcher control over when and how incident information is inspected, which is critical in multitasking environments.

The timeline does not perform interpretation or prioritization. Visual differentiation is used solely to distinguish events observed during interaction from events that occurred during dispatcher absence. Color and layout do not encode semantic meaning related to urgency, severity, or correctness.

The timeline complements the event feed and re-engagement summary by providing a persistent overview of incident evolution. While the event feed supports detailed inspection and filtering, and the summary supports orientation at the moment of return, the timeline maintains continuous access to the full temporal structure of the incident.

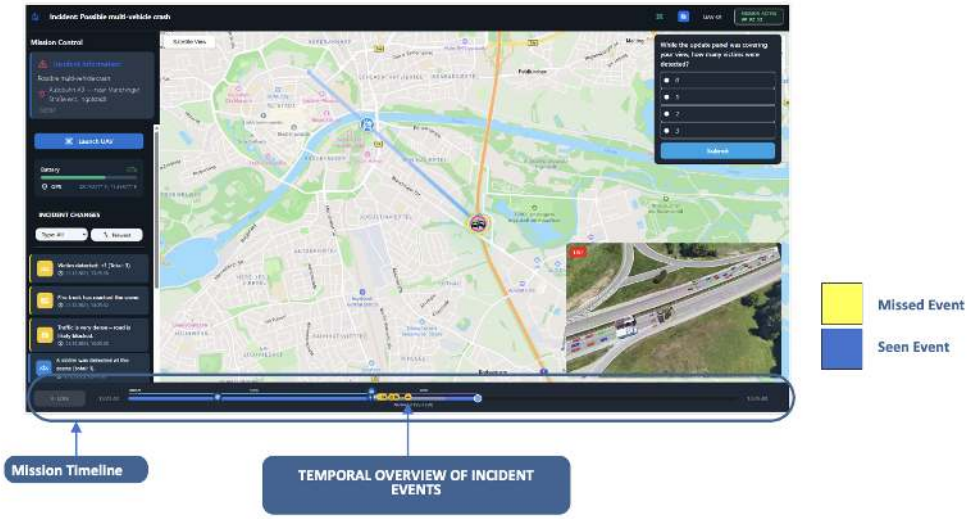

Figure 7: Timeline component showing temporally ordered UAV derived events.

4.6.4. Re-engagement summary

The re-engagement summary supports dispatcher orientation when returning to the interface after an interruption. When the dispatcher resumes interaction with an incident, the system presents a summary view that contains updates which occurred during the absence, as shown in Figure 8.

The summary is displayed as a modal overlay that appears before further interaction with the interface. This presentation was chosen to ensure that information accumulated during dispatcher absence is encountered before any further interaction with the interface. Alternative designs, such as persistent panels or optional access to summaries, would require dispatchers to actively seek out missed information and increase the risk that relevant updates remain unnoticed during re-engagement.

The timing of the summary presentation is aligned with the re-engagement phase of interaction, defined as the moment when a dispatcher returns to an incident after an interruption and reconstruct the current incident state. Presenting missed updates at this point supports orientation before further task execution and reduces reliance on memory or fragmented cues.

The content of the summary is derived from the same event data shown in the timeline and event feed. The summary aggregates incident updates that occurred during the period of dispatcher absence and presents them in a compact form. In addition to listing missed updates, simple visual indicators reflect changes in event frequency over time, such as increases or decreases in event occurrence, without providing interpretation or assessment.

The summary does not prioritize, rank, or evaluate the relevance of updates. All information remains descriptive, and assessment of importance or urgency is left to the dispatcher. This ensures that the summary supports re-engagement without embedding assumptions about dispatcher decision strategies.

The mechanisms used to detect dispatcher absence and to collect incident updates during this period are part of the system implementation and are described in the system architecture chapter.

Figure 8: re-engagement summary component supporting orientation after interruptions.

4.7. Summary

This chapter described the design process through which a dispatcher centred interface for interaction with UAV derived incident information was developed for experimental investigation. The design was grounded in an analysis of dispatcher work, interruption characteristics, and task demands, and was structured using a user centred design framework to derive explicit design requirements.

Based on these requirements, interface components were developed to support access to information accumulated during dispatcher absence, filtering of UAV derived events, and visual differentiation of incident updates. Design decisions emphasised experimental control, comparability between conditions, and avoidance of embedded decision logic in order to isolate the effects of information retrieval support.

The resulting interface conditions differ only in the availability of structured information retrieval components, while sharing identical layout, interaction structure, and visual encoding. These conditions define the experimental manipulation examined in the following chapter and provide a controlled basis for evaluating re-engagement with incident information under interruption and multitasking conditions.

5. System Architecture and Implementation

This chapter presents the system architecture and implementation structure underlying the experimental setup. It describes how the simulation environment, frontend interface, and logging components are organized, how information flows between them during an experimental trial, and how experimental conditions are technically enforced. The chapter focuses on architectural structure and control mechanisms rather than interface design or experimental results. Source code for the frontend interface and simulation environment is provided in public repositories referenced in Appendix A.11.

5.1. Purpose and system overview

This chapter describes the technical system developed to support the experimental investigation conducted in this thesis.

The system functions as a research instrument rather than an operational dispatch platform. It was designed to present simulated UAV based incident information to participants in a controlled and repeatable manner while enabling systematic manipulation of interface conditions and reliable logging of interaction data.

At the architectural level, the system separates incident generation, interface presentation, and interaction logging into distinct components. This separation enables experimental conditions to be realized through controlled system configurations while maintaining consistent task structure and interaction flow across trials.

The architecture distinguishes simulation and interaction concerns. Incident dynamics and sensor data are generated within a Unity based simulation environment, while all participant interaction takes place through a web based frontend implemented using standard web technologies. Simulation behavior remains independent of interface logic, and experimental conditions are enforced through paired frontend and simulation configurations.

Figure 9 provides a high level overview of the system architecture used in the experiment.

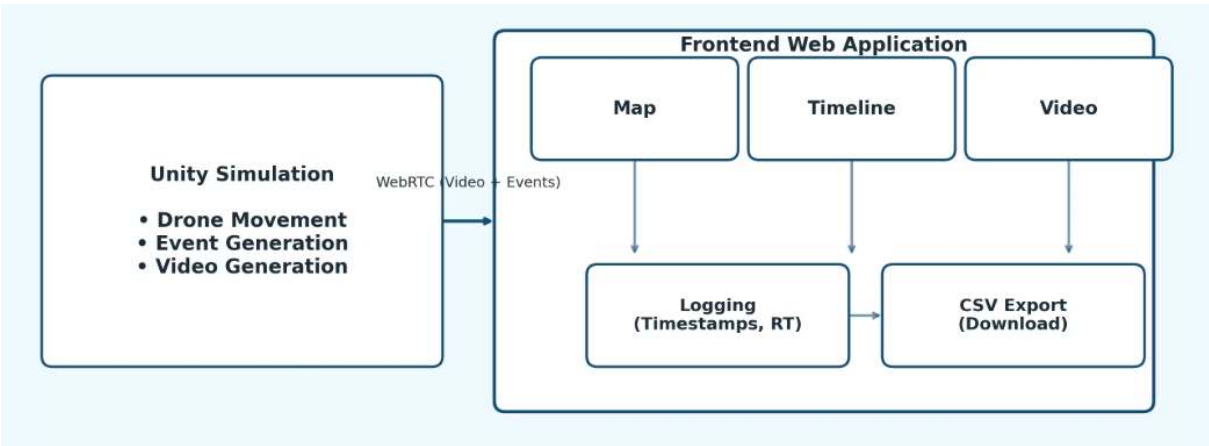

Figure 9: High level system architecture showing separation between the simulation environment, frontend web application, and interaction logging. The Unity simulation generates incident events and video streams that are presented through the frontend interface and recorded for subsequent analysis.

5.2. Information flow and experimental control

This subsection describes how incident information generated by the simulation environment is transmitted to the frontend interface during an experimental trial and how experimental control is maintained.

Incident events are generated by the simulation environment according to predefined scenario scripts and transmitted unidirectionally to the frontend interface. The frontend reflects the current simulation state but does not provide feedback that could influence event generation, timing, or simulation behavior. Participants interact exclusively with the frontend interface.

Experimental interruptions are implemented as controlled secondary tasks that divert participant attention away from the primary interaction. During these periods, the simulation continues to generate incident events, which accumulate without being actively attended to by the participant.

During interruption phases, participant interaction with the frontend interface is suspended by design as part of the experimental task flow. The start and end of each interruption are therefore known to the system. re-engagement is detected when the participant completes the secondary task and interaction with the frontend interface resumes, marking the transition from interruption to re-engagement within the system.

Upon return to the primary task, a system-mediated re-engagement phase supports orientation before interaction resumes. The manner in which accumulated information is presented during re-engagement differs depending on the interface condition.

Figure 10 illustrates the temporal structure of an experimental trial, including primary interaction, interruption, re-engagement, and continued interaction phases.

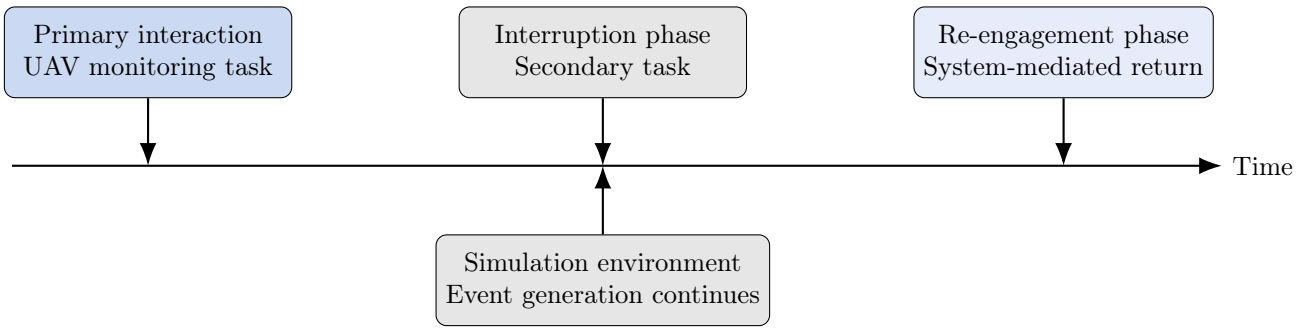

Figure 10: Temporal structure of an experimental trial. Participants perform a primary UAV monitoring task followed by an artificial interruption task. During the interruption, incident events continue to be generated by the simulation environment. Upon return, a system-mediated re-engagement phase supports orientation before interaction resumes, depending on the interface condition.

5.3. Frontend interface structure

This subsection describes the structure of the frontend interface as the sole point of interaction between participants and the experimental system.

The frontend interface is implemented as a web based application and is responsible for presenting information generated by the simulation environment. It does not perform simulation logic or influence incident dynamics. All participant interaction is limited to inspection and navigation within the interface.

The interface integrates multiple coordinated views to support access to incident information. A spatial view presents the geographic context of the incident and detected elements. A temporal view represents the sequence of events over time. An event based view provides textual descriptions of detected events. These views present the same underlying incident data in complementary forms.

Two frontend interface implementations were created corresponding to the experimental conditions. Both implementations share the same layout structure, interaction mechanisms, and

visual encoding. Differences between conditions are limited to the visibility and availability of information retrieval and re-engagement components, while the overall interface structure is preserved.

By implementing condition specific behavior at the interface level and pairing it with controlled simulation configurations, the system enables comparison of interaction behavior across conditions without altering task structure or interaction flow.

Detection of interruption boundaries and triggering of re-engagement related interface behavior are implemented at the frontend level based on the experimental task flow, while incident event generation remains fully controlled by the simulation environment.

5.4. Experimental conditions and control logic

This subsection describes how the experimental conditions are realized within the system architecture and how experimental control is maintained.

Two experimental conditions were implemented: WIR and WOIR. Each condition is realized through a paired configuration consisting of a dedicated frontend interface and a corresponding simulation scene. These configurations were designed to be scenario-equivalent while differing in the availability and presentation of incident information.

At the frontend level, the two conditions use separate interface implementations. In the WIR condition, information retrieval and re-engagement components are visible and integrated into the interface. In the WOIR condition, these components are hidden, requiring participants to rely on manual inspection of available information. Aside from the availability of these components, layout structure and interaction mechanisms remain consistent.

At the simulation level, separate Unity scenes were created for the two conditions. The scenes share the same environment layout, drone behavior, and task structure, but differ in the incident information present during the trial. These differences include the number and type of detected entities and incident events, while preserving overall scenario complexity and progression.

Scenario scripts were constructed to ensure that overall event density, timing, and task demands were comparable across conditions, thereby limiting systematic differences to the availability of information retrieval support at the interface level.

The frontend interface is connected to the simulation environment via WebRTC. Video streams, drone position updates, and incident events are generated by the simulation and transmitted to the frontend in real time. The frontend map reflects drone movement based on coordinates received from the simulation, and detected events are displayed as they occur.

By pairing frontend and simulation configurations per condition and keeping task structure, timing, and interaction flow consistent, the system enables controlled comparison of participant behavior across conditions while preserving experimental validity.

5.5. Interaction logging and data collection

This subsection describes how participant interactions and system events are recorded for subsequent analysis.

Interaction logging is implemented at the frontend interface level and captures all participant actions relevant to the experimental tasks. Logged events include navigation actions, interactions with interface components, and responses to task related prompts. Logging does not affect interface behavior or simulation execution.

In addition to participant interactions, the system records timestamps associated with incident events received from the simulation environment. This allows alignment of participant actions with the temporal progression of the simulated scenario, including periods of interruption and re-engagement.

Logging is applied consistently across experimental conditions. Both frontend implementations generate log data using the same structure and event definitions, ensuring comparability across conditions. Each log entry includes a participant identifier, condition label, event type, and timestamp.

Log data are exported in a structured format at the end of each experimental session and stored for later quantitative and qualitative analysis. No data are recorded during training phases, and logging is limited to experimental trials only. A complete summary of all logged interactions is provided as a CSV file in Appendix A.10.

5.6. Summary

This chapter described the system architecture and implementation structure developed to support the experimental investigation conducted in this thesis. The system was designed as a controlled research instrument rather than an operational dispatch platform.

The architecture separates simulation, frontend interaction, and logging components to enable controlled realization of experimental conditions. Incident dynamics and visual content are generated within a Unity based simulation environment, while participant interaction is mediated exclusively through a web based frontend interface. Interaction logging captures participant behavior and system events in a structured and comparable manner across conditions.

By realizing experimental conditions through paired frontend and simulation configurations and maintaining consistent task structure and interaction flow, the system supports systematic comparison of interaction behavior under interruption and re-engagement conditions. The following chapter describes the experimental methodology used to evaluate these conditions.

6. Methodology

This chapter describes the methodology used to evaluate the UAV interface described in Chapter 5. The study investigates how information retrieval support influences dispatcher performance, cognitive load, and SA during multitasking and interruptions. A controlled user study was conducted in a simulated emergency UAV environment to enable systematic comparison between interface conditions.

6.1. Study Design Overview

This section outlines the overall research approach, experimental design, and interface conditions used in the study. Together, these elements define how the evaluation was structured and how comparisons between interface variants were conducted.

6.1.1. Research Approach

The study follows a design oriented, mixed method research approach [28, 22]. The focus is on evaluating an interactive UAV interface concept rather than developing an operational dispatch system.

Quantitative and qualitative methods are combined to examine the effects of information retrieval support under interruption conditions. Objective measures were used to assess task performance, cognitive load, and SA, while subjective questionnaires and post study interviews were used to capture user experience, interaction strategies, and perceived support during re-engagement.

6.1.2. Experimental Design

A within subject experimental design was employed [15, 26]. Each participant completed the study using two UAV interface conditions, allowing direct comparison within individuals.

The order of interface conditions was counterbalanced to reduce learning and order effects. The design isolates the effect of information retrieval support by varying only the interface features between conditions while holding all other factors constant.

6.1.3. Interface Conditions

Two interface conditions were evaluated to isolate the effect of information retrieval support.

The first condition provided basic access to UAV video and event information without explicit support for re-engagement, filtering, or visual differentiation of events WOIR.

The second condition included re-engagement summaries, incident based filtering, and visual differentiation to support information retrieval after interruptions WIR.

Detailed descriptions of the interface components and interaction mechanisms are provided in Chapter 5.

6.2. Experimental Setup

This section describes the technical and simulated environment in which the study was conducted, including the prototype used as the experimental stimulus and the simulated UAV scenario.

6.2.1. Prototype

The study used an interactive UAV interface prototype as the experimental stimulus. The prototype was designed to support controlled evaluation of dispatcher interaction under multitasking and interruption conditions and was not intended to represent a complete operational dispatch system.

Participants interacted with two predefined interface conditions derived from the same conceptual prototype. Across conditions, task structure, interaction flow, and core interface layout were kept consistent. Differences between conditions were introduced deliberately to vary the availability of information retrieval and re-engagement support for experimental comparison.

6.2.2. Simulated UAV Environment

The study was conducted in a simulated emergency UAV environment designed to provide a controlled and repeatable evaluation setting. The simulation represents a single emergency incident monitored by a UAV that provides continuous video and event updates.

The simulated incident evolved over time independently of user interaction. Event updates and interruptions were generated consistently across participants and conditions, ensuring that differences in user performance could be attributed to interface features rather than variations in the simulation.

6.2.3. Testing Environment

The study was conducted in a controlled laboratory environment configured to support interaction with the UAV prototype and eye tracking data collection. Figure 11 shows the physical setup used during the study.

Participants were seated at a desk equipped with three displays. The central display presented the UAV prototype interface used for task execution. A separate display was used to run the eye tracking software interface, which allowed the experimenter to monitor calibration quality and recording status during the session. An additional display was present for running Unity simulation but participants need to interact only with front screen.

Eye tracking was performed using a remote, desk mounted eye tracking system positioned below the primary display. The setup included two eye tracking sensors placed symmetrically to capture binocular gaze data. Participants interacted with the system using a standard keyboard and mouse.

The seating position, screen layout, and eye tracker placement were kept consistent across all participants and conditions. Lighting conditions and room layout were controlled to minimize glare and tracking interference. This setup ensured reliable gaze data collection while maintaining a stable and repeatable testing environment across sessions.

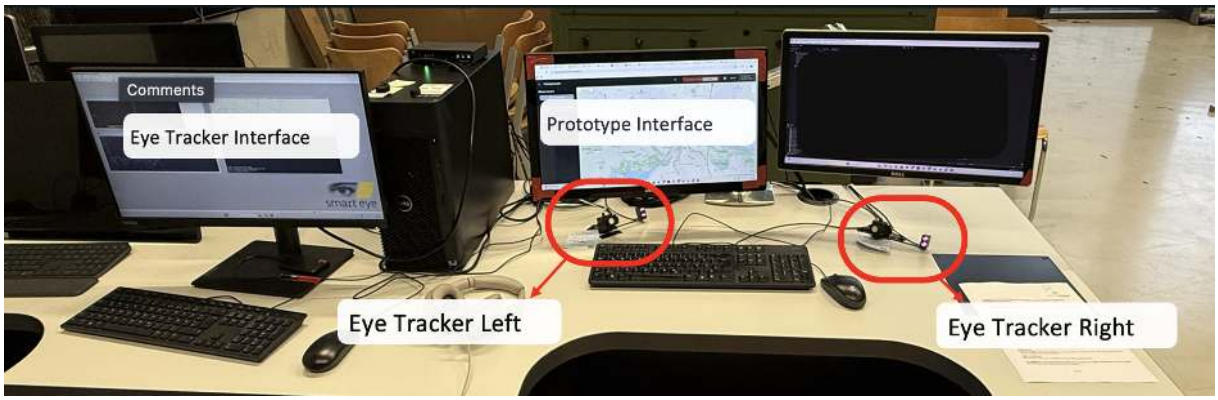

Figure 11: Physical testing environment used during the study, showing the multi display setup, eye tracking system, and participant interaction space.

6.3. Participants

A total of 29 participants took part in the study. Of these, 24 identified as male and 5 as female. Participants ranged in age from under 25 to 54 years, with the majority belonging to the 25 to 34 age group.

Most participants were students and did not have professional experience in emergency dispatching. Five participants reported prior experience as volunteer firefighters or first responders. Two participants were professional dispatchers working in an integrated control center ILS in Ingolstadt and reported between three and seven years of professional experience.

All study materials, questionnaires, and interface elements were available in both German and English. All participants completed the study successfully. An overview of participant demographics is provided in Figure 12.

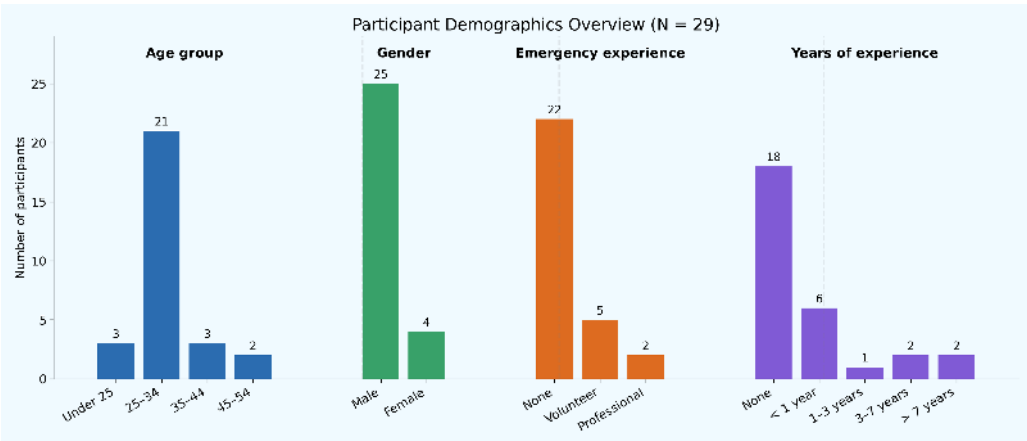

Figure 12: Overview of participant demographics, including age distribution, gender, emergency experience type, and years of experience (N = 29).

6.4. Study Procedure

The study followed a structured, within subject procedure consisting of onboarding, task execution under two interface conditions, and post-study assessment, as illustrated in Figure 13. After providing informed consent, participants completed eye-tracking calibration and a digital onboarding phase introducing the simulated emergency context, the role of the UAV, and the core interface elements. This was followed by a short familiarisation phase to ensure basic interaction competence before the main tasks.

Participants then completed the primary UAV monitoring task twice, once WIR and once WOIR, with condition order counterbalanced. During each task, participants monitored live UAV video and event updates. At predefined intervals, system-generated interruptions temporarily blocked interaction, allowing the incident to evolve in the background.

After each interruption phase, participants were presented with a set of scenario based SA probe questions. These questions assessed perception and comprehension of the evolving incident state, including number of vehicles involved, number of visible victims, presence of emergency units, traffic conditions, and hazard indicators such as smoke or fire. Participants selected their answers from predefined multiple choice options. Both response accuracy and response time were recorded to assess objective SA during re-engagement following interruption. Following completion of the probe questions, participants continued monitoring the incident under the respective interface condition (WIR or WOIR).

Following each condition, SA and usability questionnaires were administered. The session concluded with a structured post-study interview. The complete procedure lasted approximately 50 to 55 minutes per participant.

For readers interested in the detailed experimental interaction flow and interface design for both conditions, Appendix A.2 provides a visual overview.

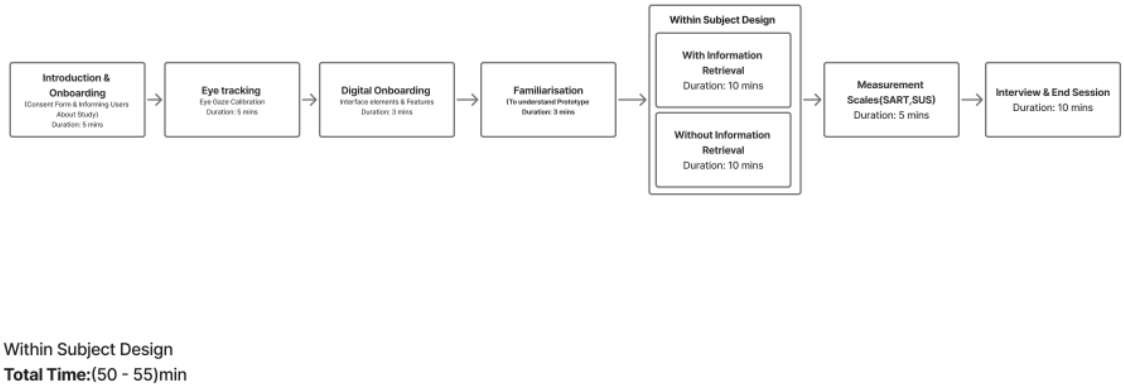

Figure 13: Overview of the within subject study procedure, including onboarding, task execution with both interface conditions, interruption phases, measurements, and post study interview.

6.4.1. Experimental Scenario and Task Flow

The experimental task simulated a single emergency incident monitored via a UAV. The interaction structure was identical across both interface conditions, with differences limited to the availability of information retrieval support during re-engagement.

Each condition began with a visual alert indicating an incoming emergency call. Participants were required to accept the call and review information provided by a simulated caller reporting a traffic accident. The initial information was intentionally incomplete to require further situational assessment during monitoring.

Participants then initiated UAV deployment using a launch control within the interface. UAV navigation and positioning were fully automated. Participants did not manually control the drone but monitored the live video stream and dynamically updated incident information.

During the monitoring phase, event updates were presented in real time. Predefined interruption periods were introduced, during which the primary interface was temporarily blocked. Participants completed short secondary tasks designed to induce cognitive disengagement from the primary monitoring activity.

A visual Detection Response Task (DRT) was presented at randomized intervals in accordance with ISO 17488. Participants responded to brief visual and auditory stimuli while continuing the primary task. The DRT ran continuously from UAV launch until completion of the Situation Present Assessment Method (SPAM) probes.

While participants were engaged in interruption tasks, the simulated incident continued to evolve. Upon returning to the interface, system state changes that occurred during absence were visible.

In the WOIR condition, participants resumed monitoring without additional re-engagement support. In the WIR condition, a re-engagement summary highlighted missed events and visually differentiated updates that occurred during absence.

Following re-engagement, participants completed a set of SPAM probe questions assessing perception and comprehension of the current incident state. After answering all probes, monitoring continued until task completion.

All user interactions, event updates, interruption timing, DRT responses, and SPAM responses were automatically recorded via the system logging framework. Figure 14 illustrates the temporal structure of the scenario.

All user interactions, event updates, interruption timing, DRT responses, and SPAM responses were automatically recorded via the system logging framework. Figure 14 illustrates the temporal structure of the scenario.

Measurement Overview

The experimental task incorporated three primary measurement components:

- **Task completion time** captured overall task efficiency from mission start until completion of the SPAM probes.
- **DRT** measured attentional demand continuously throughout monitoring and interruption phases.

- **SPAM** assessed objective SA immediately following re-engagement.

Together, these measures capture performance, cognitive load, and SA under interruption conditions.

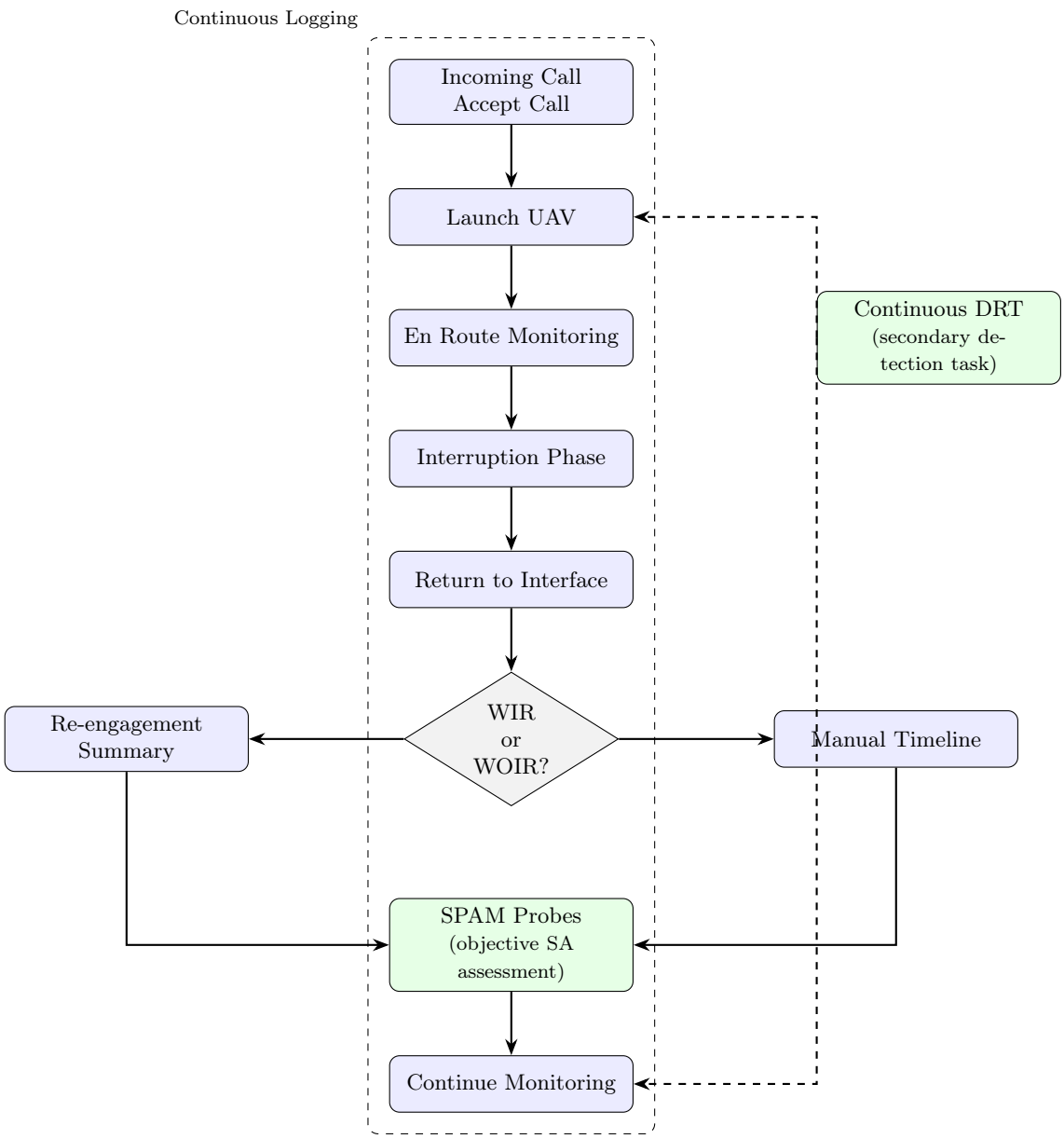

Figure 14: Experimental scenario flow including interruption phase, condition-specific re-engagement, continuous DRT, and SPAM assessment.

6.5. Data Collection

This section describes the quantitative and qualitative data collected to evaluate dispatcher performance, cognitive load, situational awareness, and user experience. Each data source was selected to capture complementary aspects of interaction with the UAV interface under multi-tasking and interruption conditions.

6.5.1. Performance Measures

Task performance was operationalized as task completion time, serving as a global efficiency measure of primary task execution. Completion time was defined as the duration between the system logged event `mission.started` and the event `mission.completed` within each interface condition.

This interval included the full monitoring sequence: answering the initial call, observing live UAV video and event updates, experiencing system-generated interruption phases, and completing the situational awareness probe questions following each interruption.

All timestamps were automatically recorded by the experimental logging system. For each participant and condition, completion time was calculated as:

$$\text{completion time} = t_{\text{mission.completed}} - t_{\text{mission.started}}$$

For statistical analysis, completion time was aggregated at the participant level as a single total duration per condition. Time was measured in seconds. Only participants with valid log data in both conditions were included in the within-subject comparison.

Lower completion times indicate higher task execution efficiency.

6.5.2. Cognitive Load Measurement

Cognitive load refers to the mental effort required to perform the primary task while managing parallel demands. In interruption prone environments, changes in interface design may influence performance by altering attentional demand rather than task difficulty itself. Cognitive load measures were therefore included to assess whether information retrieval support affects the amount of mental effort required during interaction.

Detection Response Task

Cognitive load was assessed using a visual DRT in accordance with ISO 17488 [48] and established DRT implementations in multitasking research [60]. The DRT was implemented as a secondary task running in parallel with the primary UAV monitoring task. The DRT ran continuously from UAV launch until completion of the SPAM probes.

After the incident call was taken and the mission started, brief visual stimuli were presented at random intervals between 8 and 14 seconds. Each stimulus consisted of a short visual flash accompanied by a brief auditory signal. Participants were instructed to press a designated keyboard key as quickly as possible upon detecting the stimulus.

Response time was recorded as the interval between stimulus onset and key press. If no response was registered within 2.5 seconds, the stimulus was logged as a miss.

Mean response time and miss rate were calculated per participant and condition. For inferential testing, the dependent variable reported in the results corresponds to the mean response time (in milliseconds) of valid DRT responses per participant and condition. Increased response times and higher miss rates indicate higher cognitive load during primary task execution.

6.5.3. Situation Awareness Measures

Situation awareness measures assess participants' understanding of the current system state and ongoing incident dynamics during task execution. Both objective and subjective measures were used to capture different aspects of situation awareness under interruption conditions.

SPAM

SA during task execution was assessed using SPAM probe questions [56]. In contrast to post task self report measures, SPAM captures objective situational awareness through scenario specific queries presented during task execution.

Following each interruption phase, participants were presented with a set of five multiple choice probe questions assessing perception and comprehension of the current incident state. Questions addressed incident relevant variables such as:

- number of vehicles involved,
- number of visible victims,
- presence and type of emergency units,
- road access status,
- presence of hazards such as smoke or fire.

Participants selected their responses from predefined answer options. Both response accuracy and response time were recorded. Accuracy was computed as the proportion of correctly answered SPAM probes relative to the total number of probes per participant and condition, expressed as a percentage. Response time was measured as the interval between probe onset and answer selection and was aggregated as the mean probe response time (in seconds) per participant and condition. Accuracy reflects correctness of SA, while response time reflects the accessibility of relevant incident information during re-engagement.

SART

Subjective SA was assessed using the Situational Awareness Rating Technique (SART) questionnaire [63], serving as a self reported complement to objective SPAM measures. SART captures participants' perceived understanding of the situation, attentional demand, and available mental resources following task completion, complementing objective SPAM measures.

6.5.4. Eye Tracking

Eye tracking data were collected to examine visual attention allocation during interaction with the interface [69, 45]. Gaze data were recorded continuously throughout task execution in both interface conditions.

Gaze position was sampled at a fixed rate and logged as normalized screen coordinates (x and y) relative to the primary interface display. Only gaze samples corresponding to the main experimental screen were included in subsequent analysis.

Recording began after calibration and continued throughout the monitoring task, including interruption phases and re-engagement periods. Raw gaze logs were exported per participant and condition for subsequent preprocessing and Area of Interest (AOI) based analysis.

6.5.5. System Logs

System logs were recorded to capture objective interaction data, including UAV events, user actions, interruptions, and probe responses [20]. Timestamped logs enabled synchronization across performance, workload, and situational awareness measures [20].

6.5.6. Qualitative Feedback

Qualitative feedback was collected through structured post study interviews. These interviews provided contextual insight into user experience, re-engagement strategies, and perceived strengths and limitations of the interface, supporting interpretation of quantitative results. for this

6.6. Data Analysis

This section describes the analysis strategy and statistical procedures applied to the collected data. The purpose of this section is to document the analytical approach and justify test selection. All results of these analyses are reported separately in Chapter 7.

6.6.1. Quantitative Analysis

All quantitative analyses were conducted within a within subject framework, as each participant completed both interface conditions (WOIR and WIR). Participant level summary measures were computed separately for each condition and served as the unit of analysis.

Before inferential testing, the distribution of within participant difference scores was assessed using the Shapiro–Wilk test. This procedure is recommended for small to medium sample sizes and paired designs to assess the normality assumption underlying parametric tests [46, 15].

When the normality assumption was met, paired samples t tests were used to compare interface conditions. When the assumption was violated, the nonparametric Wilcoxon signed rank test was applied. The Wilcoxon test does not assume normality and is appropriate for paired ordinal or non-normally distributed interval data [68, 7].

This analysis strategy was applied consistently across all quantitative dependent variables, including performance measures, workload measures, SA measures, and usability scores. Effect sizes were computed for all inferential tests to support interpretation and comparability across measures, following recommendations for reporting in experimental human–computer interaction research [50, 33].

6.6.2. Eye Tracking Analysis

Eye tracking data were analyzed to examine visual attention allocation during interaction with the interface. The analysis followed established practices in Human Computer Interaction (HCI) and human factors research, combining quantitative area-of-interest (AOI) based metrics with qualitative gaze visualizations [25, 4].

Areas of interest were defined a priori based on the functional structure of the interface. Three AOIs were specified:

- Events and filtering panel
- Map and incident summary region
- Live UAV video feed

AOIs were operationalized as predefined rectangular screen regions using fixed pixel boundaries corresponding to these interface components.

For each participant and interface condition, fixation time proportions were calculated as the proportion of valid fixation samples falling within each AOI relative to total valid fixation time. Fixation-based measures are commonly used as indicators of visual attention allocation and information processing during interface interaction [19, 61].

In addition, group-level gaze density heatmaps were generated to provide descriptive visualization of aggregated attention distribution and to support interpretation of the AOI-based quantitative results.

6.6.3. Qualitative Analysis

Qualitative data from post study interviews and researcher notes were analyzed using thematic analysis. This approach is widely used in HCI research to identify recurring patterns in user experiences and interaction strategies [3].

Interview transcripts were first coded using an open coding approach. Codes were iteratively refined and grouped into higher level themes based on conceptual similarity and co-occurrence. The resulting themes were used to contextualize and complement the quantitative findings reported in Chapter 7.

6.6.4. Significance Level and Reporting

All statistical tests were conducted using a predefined significance level of $\alpha = .05$. Two tailed tests were applied throughout. Exact test statistics, p values, and effect sizes are reported for all analyses to support transparency and reproducibility [9].

6.6.5. Analysis Implementation and Reproducibility

All quantitative analyses were conducted using custom analysis scripts implemented in Python. The scripts were used to preprocess raw system logs and questionnaire data, extract dependent variables, compute participant-level summary measures, perform assumption checks, conduct inferential statistical tests, and generate figures.

For all measures, raw data were cleaned and filtered prior to analysis, including the exclusion of incomplete or invalid participant data. Dependent variables were aggregated at the participant and condition level to support within- ubject comparisons. Normality of difference scores was assessed to guide the selection of parametric or nonparametric statistical tests, and effect sizes were computed for all inferential analyses.

All analysis steps were executed using deterministic scripts, and intermediate datasets and summary outputs were stored prior to statistical testing to support transparency and reproducibility.

6.7. Ethical Considerations

The study was conducted in accordance with established ethical guidelines for human–computer interaction research. Participation was voluntary, and all participants provided informed consent prior to taking part in the study. Participants were informed about the study procedure, data collection methods, and their right to withdraw at any time without penalty.

All experimental scenarios were fully simulated, and no real emergency data or operational systems were involved. Data collected during the study were anonymized and stored securely in compliance with applicable data protection regulations. The study posed no known risks to participants, and no adverse events were reported.

6.8. Threats to validity

This section outlines potential threats to the validity of the study. The purpose is to transparently document methodological limitations and constraints that may affect interpretation of the results, independent of the analytical procedures applied. The discussion addresses internal validity, construct validity, statistical conclusion validity, and ecological validity.

6.8.1. Internal validity

Internal validity concerns whether observed differences between interface conditions can be attributed to the experimental manipulation rather than to uncontrolled factors. A within subject design was used to reduce between participant variability, but such designs may introduce learning or carryover effects. Although condition order was counterbalanced, participants may still have become more familiar with the task structure, interface layout, or interruption pattern over time.

The study included a familiarisation phase to support basic interaction competence, but extended training was not provided. As a result, participant interaction strategies may have continued to evolve during task execution. In addition, interruptions were system generated and followed a predefined pattern. While this ensured consistency across participants and conditions, it may

not capture the full variability of interruption timing and content present in real dispatcher environments involving UAV based monitoring.

6.8.2. Construct validity

Construct validity refers to the degree to which the selected measures accurately represent the intended constructs. In this study, SA was assessed using both objective and subjective measures. The SPAM method captures situation related knowledge during task execution, but it does not represent all dimensions of SA, such as longer term comprehension or anticipatory understanding.

Cognitive load was assessed using a DRT, which reflects attentional demand during multitasking but does not directly measure subjective effort or perceived workload. Subjective SA was assessed using the SART questionnaire, which relies on participant self report and may be influenced by individual interpretation or response tendencies.

Eye tracking data were used to examine visual attention allocation across interface components. While gaze behaviour provides insight into where participants look, it does not directly indicate cognitive processing or understanding. Fixation based measures within predefined AOI therefore reflect attention distribution rather than decision making or comprehension.

6.8.3. Statistical conclusion validity

Statistical conclusion validity concerns the reliability of inferences drawn from the statistical analyses. The study included 29 participants, which is typical for controlled laboratory studies in HCI research but may limit sensitivity to small effects. As a result, the absence of statistically significant differences cannot be interpreted as evidence of equivalence between interface conditions.

Multiple dependent variables were analysed to capture complementary aspects of performance, cognitive load, SA, and System Usability Scale (SUS). Although consistent analysis procedures were applied across measures, the number of comparisons increases the complexity of interpretation. Effect sizes were therefore reported alongside significance values to support transparent assessment of observed differences.

6.8.4. Ecological validity

Ecological validity addresses the extent to which the study setting reflects real world dispatcher work. The study was conducted in a simulated emergency environment focusing on a single incident monitored via a UAV. Real dispatch contexts often involve multiple concurrent incidents, collaboration between operators, and organisational procedures that were not represented in the experimental setup.

Most participants were students and did not have professional dispatch experience in an ILS. While some participants reported emergency response experience, the sample does not represent the full range of expertise found in operational control centers. In addition, the study examined short term interaction during a single session. Longer term adaptation, learning effects, and sustained use were not assessed.

These limitations reflect deliberate design choices made to support experimental control and repeatability. The findings should therefore be interpreted within the context of a controlled laboratory study examining interaction with simulated UAV based incident information under interruption conditions.

6.9. Methodology Summary

This chapter described the methodological framework used to evaluate the UAV interface under multitasking and interruption conditions. A controlled within-subject user study was conducted in a simulated emergency environment to compare two interface variants with and without information retrieval support. The chapter detailed the study design, experimental setup, participant sample, study procedure, data collection instruments, and analysis strategy, establishing a transparent and reproducible basis for the results reported in Chapter 7.

7. Results

This chapter reports the results of the user study conducted to compare the interface WOIR and the interface WIR. The chapter presents quantitative and qualitative findings derived from the collected data.

Quantitative results are reported first, including task performance, cognitive workload, situational awareness, usability, and visual attention measures. Qualitative findings from post study interviews are reported last. All results are presented without interpretation. Explanatory discussion is deferred to Chapter 8.

7.1. Overview of Quantitative Results

Table 1 summarizes the quantitative results for all dependent variables. For each measure, the table reports the number of observations, mean values per interface condition, normality test results, the statistical test applied, test statistics, p values, and effect sizes. Detailed results for each measure are presented in the following subsections. Table 1 summarizes the quantitative results for all dependent variables as defined in Section 6.5.

Table 1: Complete statistical overview by dependent variable

Variable	N	WOIR	WIR	Diff	Shapiro-p	Test	p	Effect
Mean DRT response time (ms)	29	766	772	+5	.007	Wilcoxon	.624	$r = .09$
Task completion time (s)	29	262	246	−16	.168	t test	.164	$d = -.27$
SPAM accuracy (%)	31	32	48	+15	.003	Wilcoxon	**.005**	$r = .65$
Mean SPAM response time (s)	31	19.6	14.4	−5.2	.131	t test	**.006**	$d = -.54$
Demand (SART)	29	13.1	13.9	+0.8	.007	Wilcoxon	.225	$r = .61$
Supply (SART)	29	10.1	10.6	+0.5	.141	t test	.207	$d = .24$
Understanding (SART)	29	9.2	9.9	+0.7	.210	t test	.200	$d = .24$
Overall SART	29	6.2	6.5	+0.3	.155	t test	.688	$d = .08$
SUS	28	63.2	58.4	−4.8	.053	t test	.079	$d = -.35$

Note. Shapiro-p refers to the Shapiro–Wilk test on paired difference scores. WOIR = without information retrieval, WIR = with information retrieval.

7.1.1. Task Performance and Workload

This section reports results related to task performance and cognitive workload under the two interface conditions. Task performance was assessed using task completion time. Cognitive workload was assessed using detection response task reaction times. All analyses were conducted using within participant comparisons.

7.1.2. Task Completion Time

Task completion time was computed for each participant and interface condition. Data from 29 participants with complete measurements across both conditions were included in the analysis.

Mean task completion time was 261.5 s (Standard Deviation (SD) = 57.9 s) in the WOIR condition and 246.2 s (SD = 57.9 s) in the WIR condition. Descriptive statistics are reported in Table 2.

Table 2: Task completion time by interface condition

Condition	N	Mean (s)	SD (s)
WOIR	29	261.5	57.9
WIR	29	246.2	57.9

Note. Task completion time is reported in seconds.

Mean task completion times by condition are shown in Figure 15.

Normality of within participant task completion time differences was assessed using the Shapiro Wilk test and did not indicate a deviation from normality ($p = .168$). A paired samples t test was therefore applied. The test did not reveal a statistically significant difference between conditions, $t(28) = -1.43$, $p = .164$, $d = -0.27$.

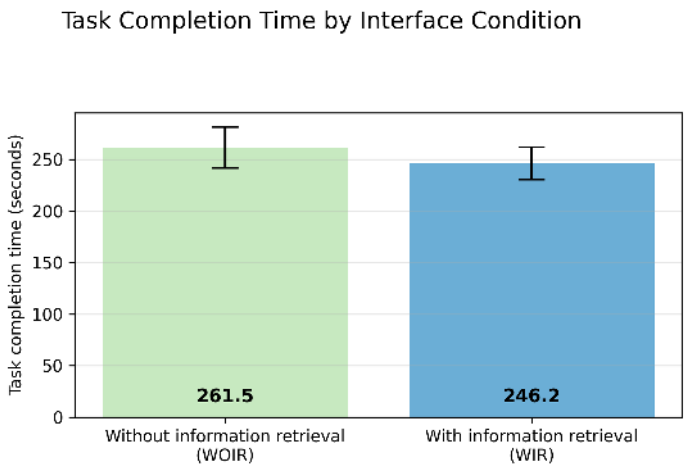

Figure 15: Mean task completion time for the interface without information retrieval (WOIR) and with information retrieval (WIR). Error bars indicate 95% confidence intervals.

Participant-level paired task completion times are shown in Appendix C, Figure 24.

7.1.3. Detection Response Task

Detection response task reaction times were analyzed for each participant and interface condition. Data from 29 participants with complete measurements across both conditions were included in the analysis.

Mean reaction time was 766.2 ms (SD = 361.3 ms) in the WOIR condition and 771.5 ms (SD = 361.3 ms) in the WIR condition. Mean reaction times by condition are shown in Figure 16.

Normality of within-participant reaction time differences was assessed using the Shapiro–Wilk test and indicated a deviation from normality ($p = .007$). A Wilcoxon signed-rank test was therefore applied. The test did not reveal a statistically significant difference between conditions, $W = 194.00$, $p = .624$, $r = .094$.

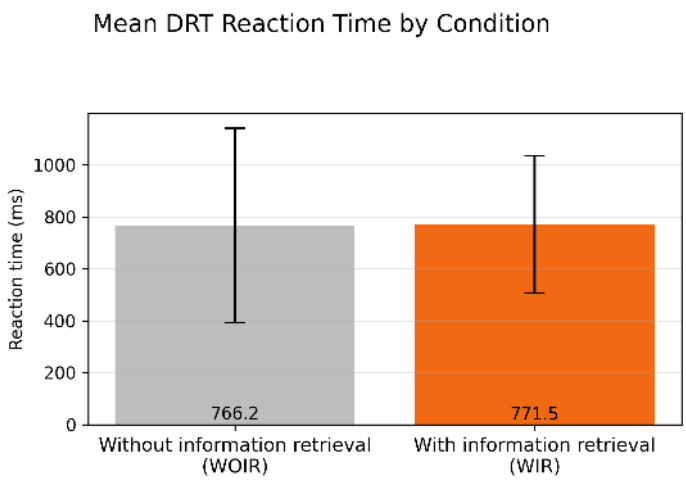

Figure 16: Mean detection response task reaction time for the interface without information retrieval (WOIR) and with information retrieval (WIR). Error bars indicate one standard deviation.

Participant-level paired detection response task reaction times and distribution diagnostics (histogram and Q–Q plot) are provided in Appendix A (Figures 27 and 28).

7.1.4. Situational Awareness

SA was assessed using objective and subjective measures. Objective situational awareness was measured using the SPAM, capturing probe accuracy and response time during task execution. Subjective situational awareness was measured using the SART questionnaire completed after each condition. Only participants with complete paired data were included in the analyses.

7.1.5. Objective Situational Awareness

Objective SA was assessed using SPAM probe accuracy and probe response time. Data from 29 participants with complete paired measurements were included.

Mean probe accuracy was higher in the WIR condition ($M = 47.58\%$, $SD = 20.77$) than in the WOIR condition ($M = 32.26\%$, $SD = 21.60$). Normality of within-participant accuracy difference scores was violated ($p = .003$); therefore, a Wilcoxon signed rank test was applied. The analysis revealed a statistically significant difference between conditions, $W = 64.50$, $\mathbf{p = .005}$, $r = .65$.

Mean probe response time was lower in the WIR condition ($M = 14.40$ s, $SD = 5.25$ s; corresponding to 14 399.21 ms) than in the WOIR condition ($M = 19.57$ s, $SD = 9.20$ s; corresponding to 19 566.44 ms). Normality of paired response time difference scores was not violated ($p = .131$), and a paired samples t test was applied. The difference between conditions was statistically significant, $t(28) = -2.98$, $\mathbf{p = .006}$, $d = -0.54$.

Descriptive statistics are reported in Table 3.

Table 3: Objective situational awareness performance measured using SPAM by condition. Response times are reported in seconds.

Condition	Accuracy Mean (%)	Accuracy SD	Incorrect Mean	Incorrect SD	RT Mean (s)	RT SD (s)
WIR	47.58	20.77	2.10	0.83	14.39	5.25
WOIR	32.26	21.60	2.71	0.86	19.56	9.20

Mean SPAM probe accuracy by condition is shown in Figure 17, and mean probe response time by condition is shown in Figure 18.

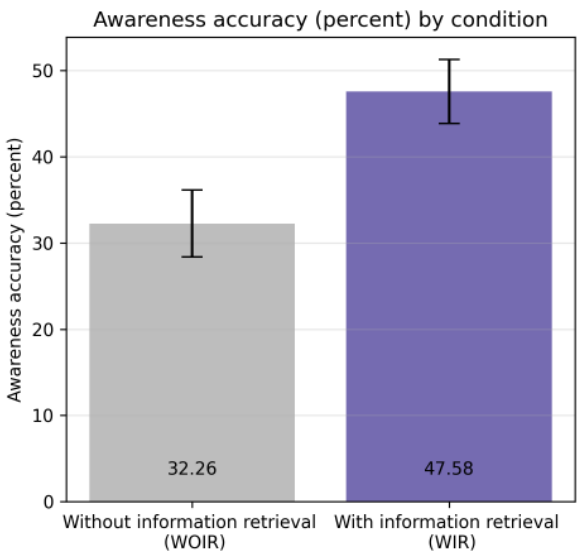

Figure 17: Mean SPAM probe accuracy by interface condition. Error bars indicate standard deviation.

Distributional diagnostics for within-participant SPAM accuracy and probe response time difference scores are provided in Appendix A (Figures 31–34).

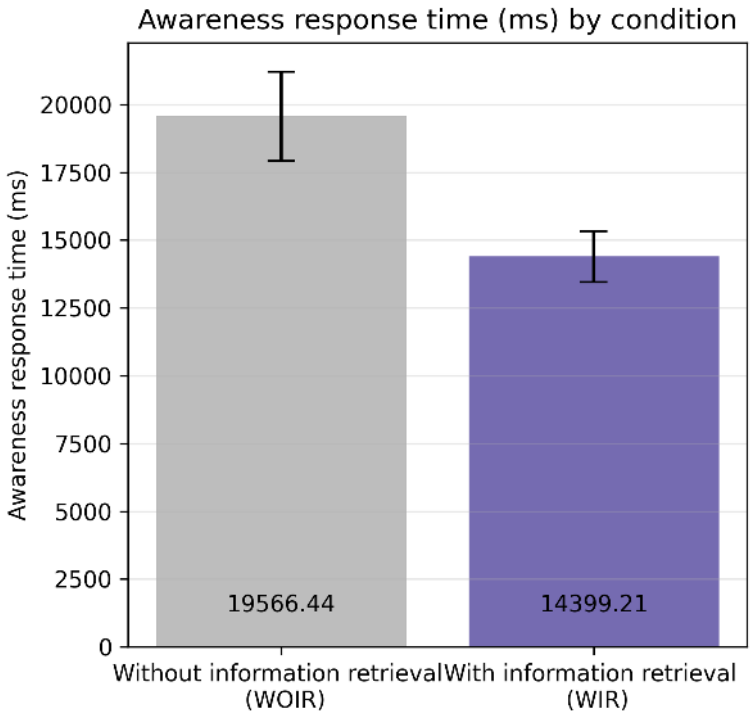

Figure 18: Mean SPAM probe response time by interface condition (milliseconds). Error bars indicate standard deviation.

7.1.6. Subjective Situational Awareness

Subjective situational awareness was assessed using the SART. Demand, Supply, Understanding, and overall SART scores were computed for each participant and condition ($N = 29$).

Mean scores were numerically higher in the WIR condition across all SART components. Normality of paired difference scores was violated for Demand ($p = .007$), and a Wilcoxon signed rank test was applied. No statistically significant difference was observed ($p = .225$).

For Supply, Understanding, and overall SART, normality assumptions were not violated (all $p \geq .14$), and paired samples t tests were conducted. None of these comparisons reached statistical significance (all $p \geq .20$).

Descriptive statistics are reported in Table 4.

Table 4: Mean SART component scores by interface condition ($N = 29$)

Component	WOIR ($M \pm SD$)	WIR ($M \pm SD$)	Test	p
Demand	$13.41 \pm SD_{\text{WOIR}}$	$14.03 \pm SD_{\text{WIR}}$	Wilcoxon	.225
Supply	$10.19 \pm SD_{\text{WOIR}}$	$10.48 \pm SD_{\text{WIR}}$	t test	.207
Understanding	$9.31 \pm SD_{\text{WOIR}}$	$9.81 \pm SD_{\text{WIR}}$	t test	.200
Overall SART	$6.09 \pm SD_{\text{WOIR}}$	$6.26 \pm SD_{\text{WIR}}$	t test	.688

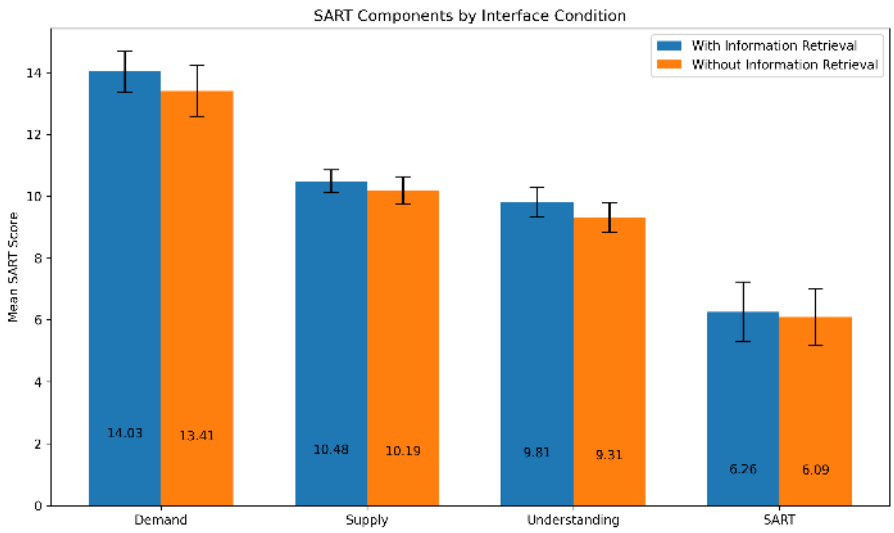

Figure 19: Mean SART component scores by interface condition. Error bars indicate standard deviation.

Participant-level paired scores and distributional diagnostics for all SART components are provided in Appendix A (Figures 35–38).

7.1.7. System Usability Scale

Perceived usability was assessed using the SUS. For each participant, SUS scores were computed separately for the WOIR and WIR interface conditions. Only participants with complete paired data were included in the analysis ($N = 29$).

Mean SUS score was 63.21 (SD = 14.94) in the WOIR condition and 57.93 (SD = 14.52) in the WIR condition. Descriptive statistics are reported in Table 5.

Normality of paired difference scores was assessed using the Shapiro–Wilk test and did not indicate a deviation from normality ($p = .053$). A paired samples t test was therefore applied. The test did not reveal a statistically significant difference between conditions, $t(27) = -1.83$, $p = .079$, $d = -0.35$.

Table 5: Mean SUS scores by interface condition ($N = 29$)

Condition	Mean	SD
WOIR	63.21	14.94
WIR	57.93	14.52

Note. SUS scores range from 0 to 100.

Mean SUS scores by condition are shown in Figure 20.

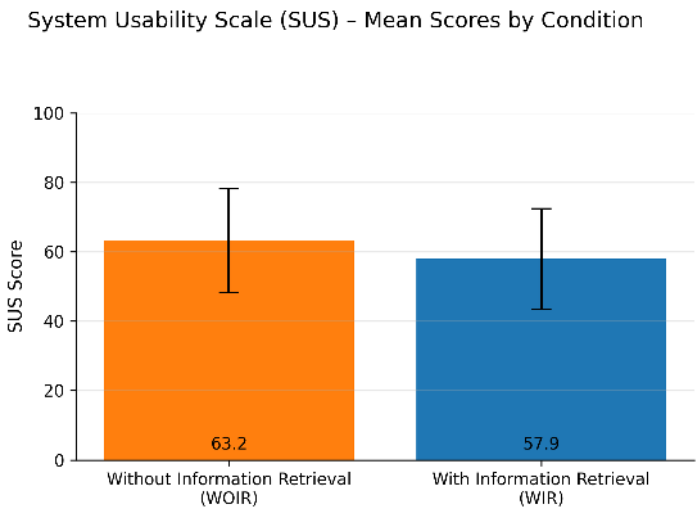

Figure 20: Mean System Usability Scale (SUS) scores by interface condition. Error bars indicate standard deviation.

Participant-level paired SUS scores and distributional diagnostics are provided in Appendix A (Figure 39).

7.2. Eye Tracking

Eye tracking results are reported based on the analysis procedure described in Section 6.6.2. Fixation time proportions within predefined AOI were compared between interface conditions using within-participant statistical tests. Detailed information on AOI definition, fixation processing, and statistical procedures is provided in Chapter 6.

7.2.1. Fixation Time Distribution Across Areas of Interest

Mean fixation time proportions for each AOI are shown in Figure 21. Across both interface conditions, the largest proportion of visual attention was allocated to the Map and Summary area.

Participants spent, on average, 50.20% (SD = 8.43) of fixation time on the Map and Summary area in the WIR condition and 46.40% (SD = 7.91) in the WOIR condition. This difference did not reach statistical significance.

The Video Feed accounted for 9.9% (SD = 3.12) of fixation time in the WIR condition and 10.2% (SD = 3.48) in the WOIR condition. No statistically significant difference was observed.

A statistically significant difference was observed for the Events Filter Bar. Participants allocated a greater proportion of fixation time to this area in the WIR condition (7.0%, SD = 2.65) compared to the WOIR condition (4.5%, SD = 1.98).

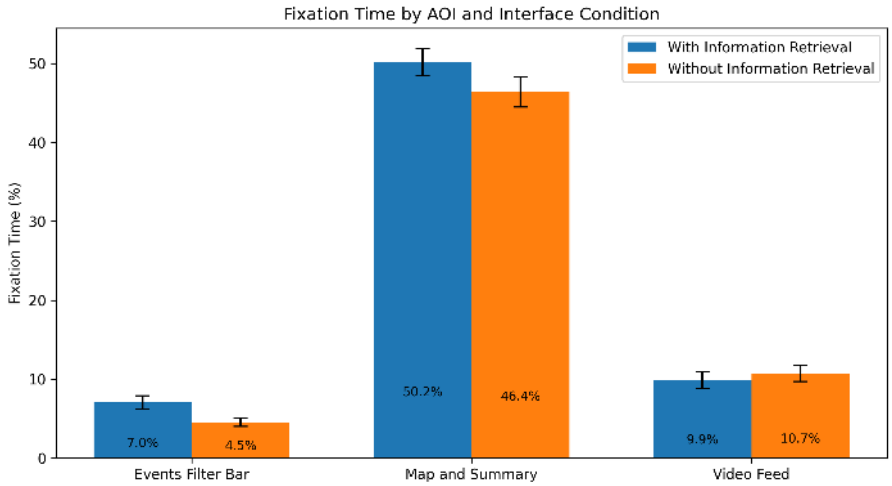

Figure 21: Mean fixation time proportions by area of interest for the interface without information retrieval (WOIR) and with information retrieval (WIR). Error bars indicate standard error of the mean.

7.2.2. Group Gaze Distribution

Figure 22 provides a descriptive overview of group-level visual attention distribution across the interface layouts for both conditions. The heatmaps visualize aggregated gaze density across participants and are included for illustrative purposes only.

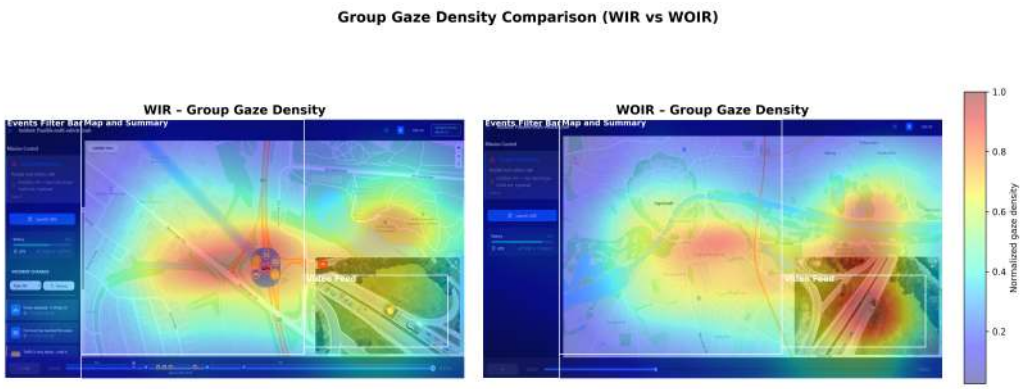

Figure 22: Group-level gaze density heatmaps overlaid on the interface layouts for the interface without information retrieval (WOIR) and with information retrieval (WIR). Heatmaps show aggregated fixation density across participants and are provided for descriptive visualization only.

7.3. Qualitative Findings

Qualitative data were collected through post study semi structured interviews conducted after participants completed both interface conditions. The analysis focused on identifying recurring

patterns in participant reports related to interaction experience, information use, and task execution. Only findings derived directly from coded interview data are reported here. Interpretation of these findings is deferred to Chapter 8.

Analysis of the interview data resulted in four main themes. These themes capture recurring aspects of participant experience across both interface conditions. Figure 23 provides an overview of the qualitative code structure used in the analysis.

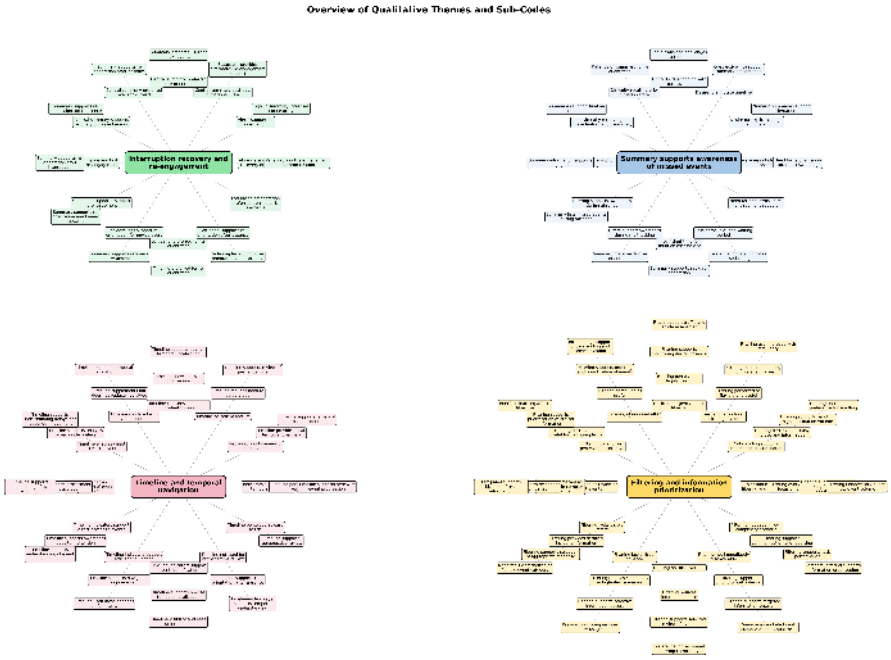

Figure 23: Overview of qualitative themes derived from post-study interviews.

7.3.1. Structured information supports re-engagement after interruptions

Participants repeatedly described structured information elements, such as summaries, event logs, and timelines, as facilitating re-engagement with the incident situation after interruptions or periods of absence. These elements were used to reconstruct changes that occurred during absence. At the same time, some participants described moments of disorientation upon returning or expressed a desire for clearer comparison between the previous and current situation state.

Sub codes associated with this theme are shown in Appendix A (Figure 40).

> "When I did not have the logs, I felt a little lost when I came back." (P01)

> "When I was not there, the system listed what happened, and that summary was more helpful when I returned." (P11)

> "You just have everything at a glance, what happened. Even if you were busy with something else." (E1)

> "While you were away it was very disorienting, but the system told me what happened." (P12)

> "It would be nice to have a delta to show the situation before I went away, because I couldn't really remember what it was like before." (P09)

7.3.2. Structured information supports situational awareness without continuous video monitoring

Participants described differences in how they monitored the incident depending on the availability of structured information. Several participants reported relying primarily on the live video feed when structured information was absent, whereas logs, alerts, and timelines were referenced when structured information was available. At the same time, some participants reported limited use of structured elements and continued reliance on the video feed even when additional information sources were present.

Sub codes associated with this theme are shown in Appendix A (Figure 41).

> "As I told, I forgot to use this function, because mostly I focused on the video." (P06)

> "The timeline gives a good overview of what is happening." (P17)

> "It's better if the information is on the page, actually, and not only in the video." (E2)

> "I never paid attention to the alerts on the left side, I think." (P10)

7.3.3. Information richness increases perceived cognitive load during early interaction

Participants reported increased mental demand when interacting with the information rich interface, particularly during early use or when multiple updates occurred simultaneously. These experiences were described as overload, difficulty distributing attention across interface elements, and challenges managing concurrent task demands. Some participants indicated that this increased demand was associated with unfamiliarity and decreased with repeated exposure.

Sub codes associated with this theme are shown in Appendix A (Figure 42).

> "I felt overwhelmed even though I understood what kind of information it was." (P01)

> "There was too much going on in the center screen, and the questionnaires popped up way too fast." (P10)

> "There was so much to see for me that I didn't even notice that the information was on the left side." (E1)

> "I haven't used the timeline; instead I used the camera to analyse the situation." (P29)

> "I will need a bit of training over how it's really structured, but once I do it a couple of times I will be more familiar." (P08)

7.3.4. Filtering, prioritization, and visual cues support information management

Participants referred to filtering mechanisms, prioritization options, and visual cues when describing how they managed the volume of information presented by the interface. These elements were mentioned in relation to selecting relevant events and maintaining awareness of incident developments. At the same time, some participants reported limited use of filtering functions or difficulty noticing them during task execution.

Sub codes associated with this theme are shown in Appendix A (Figure 43).

> "The filtering was easy to use, and I could see which responder arrived first and at what time." (P26)

> "The left panel helped me recognize how many victims were detected and who came first." (P13)

> "The change of color and the use of icons helped in identifying vehicles and victims." (P27)

> "I didn't use the filter in this scenario, but if the situation becomes more complex, it's a good thing to have." (P17)

> "I tried the filtering once and nothing happened, so I didn't pay attention to it anymore." (P10)

7.4. Summary of Results

This chapter reported the quantitative and qualitative results of the user study comparing the WOIR and WIR interface conditions.

Quantitative analyses showed no statistically significant differences between conditions for task completion time, detection response task reaction time, subjective situational awareness, or perceived usability. Statistically significant differences were observed for objective situational awareness as measured using SPAM probe accuracy and probe response time. Eye tracking analyses reported differences in fixation time allocation across areas of interest, including a statistically significant difference for the Events Filter Bar.

Qualitative analysis of post study interviews resulted in four themes describing participant reports related to re-engagement after interruptions, monitoring of incident information, perceived cognitive load during early interaction, and information management through filtering, prioritization, and visual cues. All qualitative findings were derived from coded interview data and are reported without interpretation.

Interpretation of the results and their relation to the research questions are presented in Chapter 8.

8. Discussion

This chapter discusses the findings of the study in relation to the hypotheses introduced in Section 3.8. Quantitative and qualitative evidence is integrated to explain why individual hypotheses were supported or not supported. The discussion focuses on interpretation rather than repetition of procedures or statistical results. Implications for system design are addressed after the interpretation of the findings.

8.1. Task performance

The results did not show a statistically significant effect of information retrieval support on dispatcher task completion time. Although mean task completion time was numerically lower in the WIR condition, this difference did not reach statistical significance. Accordingly, Hypothesis **H1**, which proposed higher task performance in the WIR condition (see Section 3.8), was not supported.

One explanation for this outcome is that task completion time reflects only a limited aspect of dispatcher performance in multitasking contexts. Participants were required to complete the same set of task steps in both interface conditions. Structured information retrieval supported access to information but did not reduce the number or sequence of required actions. As a result, observable differences in task duration were limited.

Qualitative findings provide additional context for this result. Participants reported using logs, summaries, and timelines primarily to regain context after interruptions rather than to complete tasks more quickly. In the absence of information retrieval support, participants relied on continuous video monitoring or memory to maintain awareness, which may have compensated for missing structured information without affecting overall task duration.

These findings indicate that Hypotheses **H1** was not supported and that task completion time alone may not capture the effects of information retrieval support in interruption prone dispatcher work.

8.2. Re-engagement and situational awareness

Clear differences between interface conditions were observed for objective SA. Participants using the WIR interface demonstrated higher SA accuracy and shorter response times compared to the WOIR interface. These findings support Hypotheses **H2a** and **H2b**, which addressed re-engagement and objective SA (see Section 3.8).

The observed pattern indicates that structured information retrieval supported access to incident information following interruptions. The SPAM probes required participants to retrieve specific system state information at particular moments during task execution. Higher accuracy and faster responses therefore indicate more direct access to relevant information when summaries, event logs, and timelines were available.

Eye tracking results are consistent with this interpretation. Participants in the WIR condition allocated a greater proportion of visual attention to the Events Filter Bar, while attention to the map and video feed remained comparable between conditions. This distribution suggests that structured interface elements were used as information access points during re-engagement.

Qualitative findings further explain how this support was used in practice. Participants reported that logs and summaries helped them identify changes that occurred during their absence, reducing the need to reconstruct incident state from memory or continuous video monitoring. In contrast, participants in the WOIR condition described difficulty identifying changes that occurred while their attention was directed elsewhere.

Taken together, the quantitative and qualitative findings indicate that Hypotheses H2a and H2b were supported and that information retrieval functionality played a central role in supporting re-engagement and objective SA.

8.3. Re-engagement as a cognitive process

The benefits observed for the WIR condition can be interpreted by viewing re-engagement as a cognitive process rather than as a general performance improvement. In interruption prone work, re-engagement requires dispatchers to rebuild their understanding of the incident after a period of absence. This involves recalling previously seen information, identifying changes that occurred during the interruption, and integrating new information into an updated mental model.

In this study, re-engagement demands were introduced through controlled interruptions while incident events continued to evolve. In the WOIR condition, participants relied on memory, continuous video monitoring, or manual inspection of past events to restore context. This form of reconstruction places demands on attention and working memory and can delay access to relevant information after an interruption.

In contrast, the WIR condition supported re-engagement through structured information retrieval. Summaries, event logs, and timelines provided representations of incident updates that occurred during participant absence. Rather than relying solely on memory or continuous video monitoring, participants could access representations of past events when resuming interaction.

Although the system did not explicitly detect interruptions, the presence of persistent structured representations supported an interruption specific workflow. When participants returned to the task, these interface elements provided an entry point for reconstructing incident state. This suggests that interfaces that are sensitive to interruption contexts could formalize such workflows, for example by highlighting changes that occurred during absence.

The improvement in objective SA observed in the WIR condition is consistent with more efficient re-engagement. Higher SA accuracy and faster response times indicate that relevant information was available at the moment it was required. Eye tracking results showing increased visual attention to interface elements associated with information retrieval during re-engagement phases support this interpretation.

Both interface conditions included identical interruption structures. Differences between conditions did not emerge in overall task completion time or subjective workload. Instead, differences were observed in objective SA accuracy and probe response time in the WIR condition.

This pattern suggests that structured retrieval support did not alter overall task duration or perceived workload. The observed differences are associated with SA measures following interruptions. If general task execution had been affected, differences in total completion time or workload ratings would be expected. The absence of such differences suggests that the effect is related to the resumption phase rather than to uninterrupted task execution.

Re-engagement therefore represents a distinct interaction phase within the task structure that depends on how historical incident information is represented.

Overall, the findings indicate that structured information retrieval in the WIR condition is associated with improved SA measures following interruptions. This supports treating re-engagement as a design consideration in dispatcher centered interfaces, separate from general task efficiency or workload reduction.

8.4. Workload and usability

The results did not show a statistically significant effect of information retrieval support on cognitive workload or perceived usability. Cognitive workload, measured using DRT, did not differ reliably between interface conditions. Perceived usability, measured using SUS, also did not show a statistically significant difference. Accordingly, Hypotheses **H3a** and **H3b** (see Section 3.8) were not supported.

The absence of increased DRT reaction times indicates that the additional information provided by the WIR interface did not introduce measurable cognitive demand under the study conditions. This suggests that structured information retrieval can be integrated without increasing workload when information is organized and retrievable.

At the same time, the absence of higher SUS scores indicates that subjective perception did not change in line with objective performance improvements. Qualitative findings provide context for this pattern. Several participants reported uncertainty during early interaction with the information retrieval interface, indicating that familiarity and learning may have influenced usability judgments during short term exposure.

This dissociation between objective measures and subjective ratings indicates that improvements in information access do not necessarily translate into immediate changes in perceived usability or workload.

8.5. Integration of quantitative and qualitative findings

Across the evaluated hypotheses, a consistent pattern emerges. Information retrieval support did not affect task completion time or subjective ratings but improved objective SA and supported re-engagement following interruptions. Quantitative analyses captured changes in information access and visual attention, while qualitative findings described how participants used structured interface elements during task execution.

The integration of findings indicates that the primary effect of information retrieval support lies in enabling access to relevant information at the moment of re-engagement rather than in altering overall task duration or perceived effort.

8.6. Implications for system design

Based on the quantitative and qualitative findings, several design implications can be derived. First, dispatcher interfaces should explicitly support re-engagement after interruptions, as structured summaries supported faster orientation following periods of absence. Second, the findings indicate that continuous video monitoring alone is insufficient for maintaining awareness across

interruptions, and that structured representations such as event logs and timelines support access to incident information. Third, filtering and prioritization mechanisms can help manage information volume without increasing measured cognitive workload. Finally, the results suggest that information richness should be balanced carefully, as multiple concurrent interface elements may increase initial interaction effort during early use.

8.7. Limitations

The findings of this study should be interpreted in light of several limitations related to the study context, participant sample, exposure duration, and scope of evaluated tasks and measures. These limitations do not invalidate the results but constrain the extent to which conclusions can be generalized beyond the experimental setting.

8.7.1. Participant sample

The participant sample consisted primarily of students, with only a small number of participants reporting professional dispatch experience in an ILS. While all participants completed the same tasks under controlled conditions, differences in prior domain knowledge, operational routines, and real world responsibility may influence how information retrieval support is used in practice.

As a result, the observed improvements in re-engagement and objective SA should be interpreted as indicative of interaction effects rather than as representative of professional dispatcher performance. The findings demonstrate how structured information retrieval can support re-engagement under interruption conditions, but they do not allow conclusions about performance or behaviour in expert populations.

8.7.2. Simulation based evaluation

The study was conducted in a fully simulated emergency environment using a single UAV monitored through an experimental interface. While this setup enabled controlled manipulation of interruptions and consistent event progression across participants, it does not capture the full complexity of operational dispatch settings.

Real dispatch environments involve multiple concurrent incidents, coordination with other operators, communication with external stakeholders, and organisational procedures that were not represented in the study. The findings therefore reflect interaction with simulated UAV based incident information rather than comprehensive dispatcher work. Conclusions should be limited to the effects of information retrieval support within controlled, single incident scenarios.

8.7.3. Short term exposure and learning effects

The study examined short term interaction during a single experimental session. Participants had limited time to familiarise themselves with the interface and to develop stable interaction strategies. As a result, subjective ratings of usability and workload may have been influenced by initial uncertainty or novelty effects.

Longer term use may lead to different interaction patterns, increased efficiency, or changes in subjective perception as familiarity increases. The present findings therefore reflect early stage interaction with information retrieval support and do not capture longer term adaptation, learning, or sustained use.

8.7.4. Scope of tasks and measures

The evaluation focused on information access, re-engagement, and SA during UAV based monitoring under interruption conditions. Task completion time, cognitive workload, and SA were assessed using established measures, but the study did not evaluate decision quality, coordination, or operational outcomes.

In addition, performance measures captured task execution characteristics rather than downstream effects on dispatch decisions. The results therefore indicate how information retrieval support influences access to incident information and re-engagement processes, but they do not support conclusions about decision effectiveness or operational impact.

Taken together, these limitations indicate that the findings should be interpreted within the scope of a controlled laboratory study examining re-engagement with simulated UAV based incident information. The results provide evidence for the role of structured information retrieval in supporting re-engagement after interruptions, while leaving broader questions of long term use, professional practice, and operational deployment to future work.

8.8. Summary

This chapter interpreted the findings of the study in relation to the proposed hypotheses and integrated quantitative and qualitative evidence. The results indicate that structured information retrieval support did not improve overall task completion time or subjective workload but significantly enhanced objective situational awareness and re-engagement following interruptions. Hypotheses **H2a** and **H2b** were supported, whereas Hypotheses **H1**, **H3a**, and **H3b** were not supported.

The findings suggest that the primary contribution of structured information retrieval lies in supporting re-engagement as a distinct cognitive phase within interruption-prone dispatcher work, rather than in increasing general task efficiency. Improvements were observed specifically in measures associated with access to incident information after interruptions, indicating that structured representations such as logs, summaries, and timelines facilitate reconstruction of incident state.

Design implications highlight the importance of supporting interruption-sensitive workflows in dispatcher-centred UAV interfaces. At the same time, the results should be interpreted within the constraints of a controlled laboratory study with short-term exposure and a primarily student sample.

Overall, the chapter demonstrates that structured information retrieval enhances objective situational awareness during re-engagement without increasing measurable workload, providing a foundation for future research on interruption-aware interface design in operational contexts.

9. Conclusion

This thesis investigated how structured information retrieval support influences dispatcher interaction with UAV based incident information under multitasking and interruption conditions. Emergency dispatchers operate in environments characterized by frequent task switching, parallel demands, and interruptions that require continuous reconstruction of incident state. Supporting effective re-engagement with ongoing incidents therefore represents a central challenge for dispatcher centered system design.

To address this challenge, a dispatcher centered interface was designed to support structured access to UAV derived incident information. The design focused on summaries, event logs, filtering, and temporal representations intended to support re-engagement after interruptions. The interface was evaluated using a controlled within subject user study comparing a condition with information retrieval support to a baseline condition without such support. The evaluation combined quantitative performance and workload measures, eye tracking data, and qualitative interview findings to capture complementary aspects of interaction.

The results showed that information retrieval support did not alter overall task completion time or subjective ratings of workload and usability. At the same time, objective measures indicated improved access to incident information when structured retrieval support was available. Participants demonstrated higher objective SA and faster responses to situational probes following interruptions in the information retrieval condition. Qualitative findings further showed that participants relied on structured interface elements to reengage with incidents after periods of absence, while also reporting uncertainty during early interaction with the system.

Taken together, these findings indicate that the primary effect of information retrieval support lies in how dispatchers access and reconstruct incident information after interruptions rather than in overall task speed or perceived usability. Structured information presentation supported objective SA during re-engagement without increasing measured cognitive workload within the controlled study setting.

9.1. Contributions

This thesis makes several contributions to research on dispatcher centered systems and interaction with UAV based incident information. First, it presents the design and implementation of a dispatcher centered information retrieval interface developed specifically to support re-engagement after interruptions. The interface was conceived as a research instrument rather than an operational system and was designed to isolate the effects of structured information access under controlled conditions.

Second, the thesis provides empirical evidence that structured information retrieval can support objective SA during interruption prone interaction without increasing measured cognitive workload. By combining performance based situational awareness measures with eye tracking and qualitative data, the study demonstrates how information retrieval mechanisms influence access to incident information at critical moments following interruptions.

Third, the thesis contributes methodologically by integrating quantitative, visual attention, and qualitative analyses within a single experimental framework. This mixed method approach enabled triangulation of findings and provided insight into not only whether differences occurred

between interface conditions, but also how participants interacted with structured information during re-engagement.

9.2. Scope of conclusions

The conclusions drawn in this thesis are bounded by the experimental design and the controlled evaluation context. The findings reflect interaction with simulated UAV based incident information in a laboratory environment and do not constitute claims about operational performance in real dispatch settings. Improvements observed for objective SA should therefore be interpreted as evidence of interaction effects under controlled conditions rather than as indicators of system effectiveness in practice.

The study examined short term interaction with a specific set of interface components and tasks. As a result, the conclusions are limited to the effects of structured information retrieval on re-engagement and access to incident state information within the defined task scope. The absence of effects on task completion time and subjective workload further indicates that information retrieval support influences specific phases of interaction rather than overall task execution.

9.3. Future work

Building on the findings and limitations of this study, several directions for future research can be identified. Future studies should evaluate structured information retrieval support with professional dispatchers in more representative laboratory or operational settings. Such studies would enable examination of how domain expertise and organizational context influence re-engagement strategies.

Longitudinal studies are needed to investigate learning effects, trust development, and changes in interaction patterns over extended use. Future work could also extend the system to additional incident types, more complex scenarios, and collaborative settings involving multiple dispatchers. Incorporating outcome measures related to decision quality, coordination, and error recovery would further broaden understanding of the role of information retrieval in dispatcher centered systems.

9.4. Final synthesis

This thesis examined how structured information retrieval affects dispatcher interaction with UAV based incident information under multitasking and interruption conditions. The findings indicate that information retrieval support primarily influences re-engagement and access to incident state information rather than overall task speed or perceived workload. Within the controlled setting of this study, structured representations such as summaries, logs, and timelines supported objective SA following interruptions without introducing additional cognitive demand.

These results highlight the importance of treating interruption recovery as a distinct interaction phase in dispatcher centered system design. By focusing on how information is accessed and reconstructed after periods of absence, rather than solely on task efficiency or automation, this thesis contributes to a more nuanced understanding of interaction with UAV based incident information under real world constraints.

Acronyms

AOI	Area of Interest	35, 36, 38, 45
CAD	Computer Aided Dispatch	4, 14
COVID19	Coronavirus Disease 2019	11
DRT	Detection Response Task	xxvii, 31–33, 38, 52
EMS	Emergency Medical Services	4, 14
HCI	Human Computer Interaction	36, 38
HSI	Human System Integration	7
ILS	Integrated Control Center	4, 9, 14–16, 29, 38, 53
SA	Situational Awareness	1–7, 12, 15, 16, 26, 27, 30, 32, 34, 35, 38, 41, 42, 50–56
SART	Situational Awareness Rating Technique	xviii–xxi, 34, 38, 41, 43, 44
SD	Standard Deviation	40–42, 44
SPAM	Situation Present Assessment Method	xv–xvii, 31–34, 38, 41–43, 49, 50
SUS	System Usability Scale	38, 44, 45, 52

UAV	Unmanned Aerial Vehicle	iii, xxvii, xxviii, 1–4, 7–9, 12, 15–18, 22–24, 26–28, 30–33, 35, 38, 39, 53–56
WIR	With Information Retrieval	xxvi, xxvii, 13, 18, 25, 27, 30–32, 35, 39–45, 49–52
WOIR	Without Information Retrieval	xxvi, xxvii, 13, 18, 19, 25, 27, 30–32, 35, 39–45, 49–51

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A. Appendices

A.1. Additional Task Performance Analyses

A.1.1. Task Completion Time

Table 7 reports participant-level mission start times, end times, and derived task completion times for each interface condition. Mean task completion times per condition are summarized at the bottom of the table.

Table 7: Participant-level mission start time, end time, and task completion time for each interface condition. Mean completion times per condition are reported at the bottom of the table.

Participant	Condition	Start time (s)	End time (s)	Completion time (s)
P01	WIR	20.77	241.29	220.51
P01	WOIR	20.89	277.16	256.27
P02	WIR	152.33	388.21	235.87
P02	WOIR	25.34	297.64	272.30
P03	WIR	27.97	237.80	209.82
P03	WOIR	22.06	237.65	215.59
P04	WIR	116.81	278.59	161.78
P04	WOIR	26.35	268.48	242.14
P05	WIR	190.96	400.11	209.15
P05	WOIR	21.80	300.80	279.00
P06	WIR	25.62	258.63	233.01
P06	WOIR	38.90	334.84	295.94
P07	WIR	22.13	272.89	250.76
P07	WOIR	24.48	315.45	290.98
P08	WIR	32.66	302.81	270.14
P08	WOIR	20.60	414.91	394.31
P09	WIR	23.45	218.86	195.41
P09	WOIR	25.35	298.04	272.69
P10	WIR	21.25	212.80	191.55
P10	WOIR	22.67	280.36	257.70
P11	WIR	27.88	280.58	252.70
P11	WOIR	24.08	333.07	308.99
P12	WIR	114.73	278.32	163.58
P12	WOIR	33.31	303.42	270.11
P13	WIR	50.00	283.31	233.32
P13	WOIR	23.04	224.61	201.56
P14	WIR	21.62	350.36	328.74
P14	WOIR	21.68	298.13	276.44
P15	WIR	24.20	302.80	278.61
P15	WOIR	64.24	374.71	310.46
P16	WIR	20.60	268.55	247.95
P16	WOIR	33.94	245.42	211.48
P17	WIR	24.93	367.00	342.06

Participant	Condition	Start time (s)	End time (s)	Completion time (s)
P17	WOIR	22.48	332.01	309.54
P18	WIR	132.41	384.88	252.46
P18	WOIR	21.70	243.95	222.25
P19	WIR	19.92	258.06	238.14
P19	WOIR	63.08	266.19	203.11
P20	WIR	19.22	272.88	253.66
P20	WOIR	22.04	409.89	387.85
P22	WIR	23.11	226.96	203.86
P22	WOIR	207.77	388.43	180.66
P23	WIR	22.43	219.35	196.92
P23	WOIR	21.74	285.57	263.83
P24	WIR	22.14	254.92	232.78
P24	WOIR	18.86	244.24	225.38
P25	WIR	28.67	293.05	264.38
P25	WOIR	22.86	317.60	294.74
P26	WIR	25.21	256.79	231.58
P26	WOIR	115.09	311.41	196.31
P27	WIR	164.56	475.22	310.66
P27	WOIR	20.33	272.89	252.56
P28	WIR	26.56	299.94	273.38
P28	WOIR	85.36	344.48	259.11
P29	WIR	180.09	486.56	306.46
P29	WOIR	22.97	313.35	290.38
P30	WIR	23.88	331.32	307.45
P30	WOIR	25.01	247.24	222.23
P31	WIR	241.80	480.55	238.75
P31	WOIR	27.32	227.68	200.37
P32	WIR	40.41	312.38	271.97
P32	WOIR	23.63	276.23	252.60
Mean completion time (s)				
Mean	WOIR	–	–	261.80
Mean	WIR	–	–	245.40

The complete participant-level datasets used for task completion time, DRT reaction time, and situational awareness analyses are provided as a supplementary spreadsheet file.

The dataset is available at: Supplementary Excel dataset (task performance data)

Figure 24 visualizes participant-level task completion times.

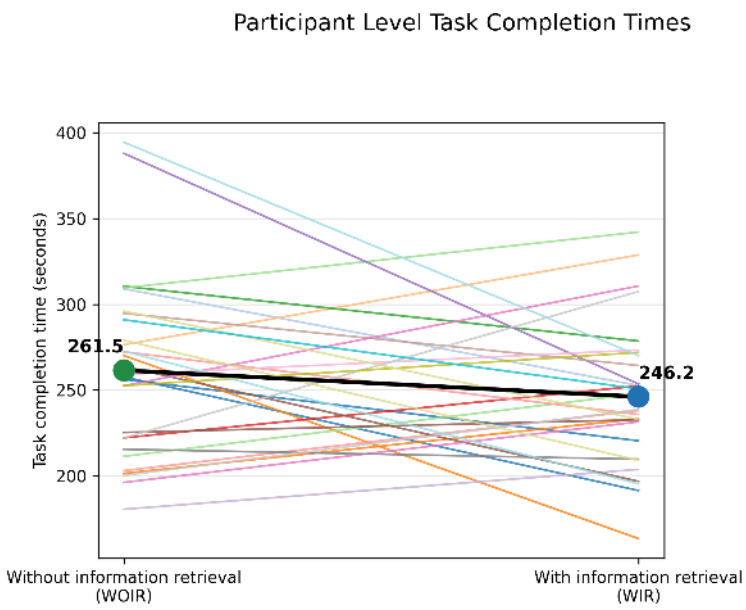

Figure 24: Participant-level task completion times for the interface without information retrieval (WOIR) and with information retrieval (WIR). Each line connects the two conditions for an individual participant.

A.2. Experimental Interaction Flow and Interface Screens

This appendix provides a visual overview of the experimental interaction flow and the corresponding interface states for both experimental conditions. The figures illustrate the sequence from UAV launch, task execution, interruption through a distraction task, and subsequent task resumption.

A.3. Information Retrieval Supported Condition (WIR)

A.4. Without Information Retrieval (WOIR)

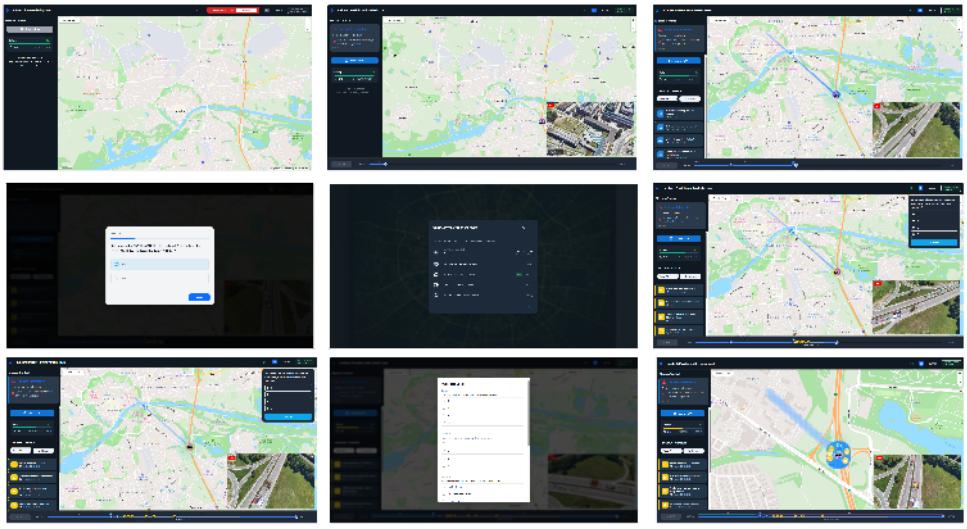

Figure 25: Experimental interaction flow and interface states in the information-retrieval-supported condition (WIR), including UAV launch, interruption, re-engagement dialog, and situational awareness probe.

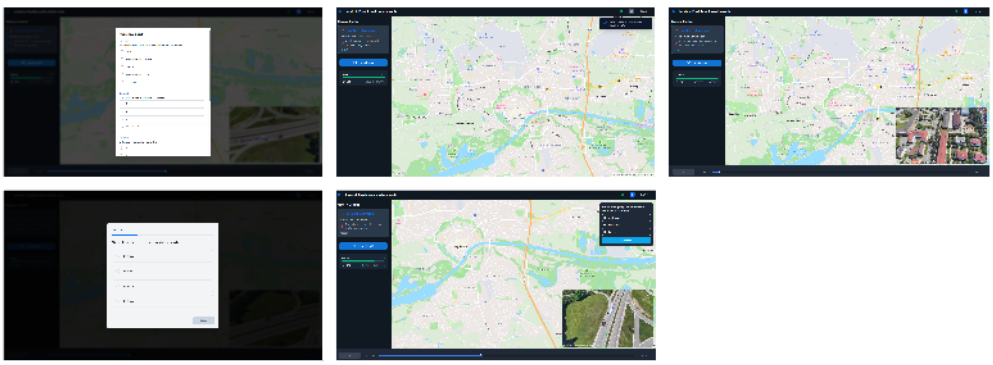

Figure 26: Experimental interaction flow and interface states in the condition (WOIR). No re-engagement dialog or additional visual incident summaries are provided after interruptions.

A.5. Statistical Assumption Checks

This section provides visual diagnostics used to assess distributional assumptions for within-participant difference scores.

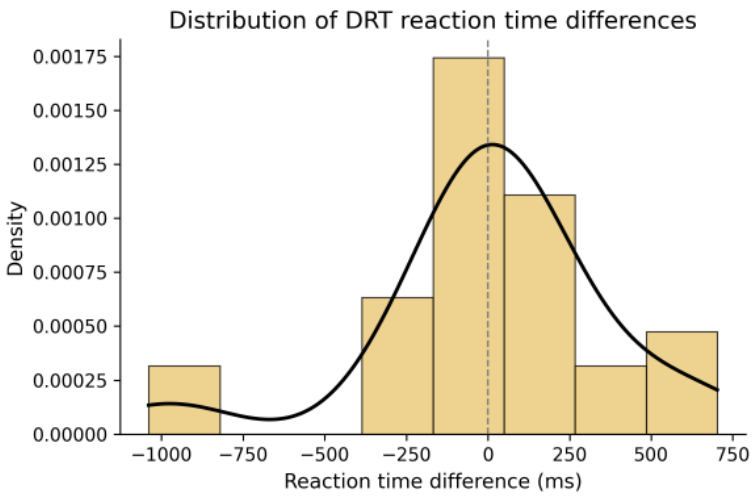

Figure 27: Histogram of within-participant detection response task reaction time differences.

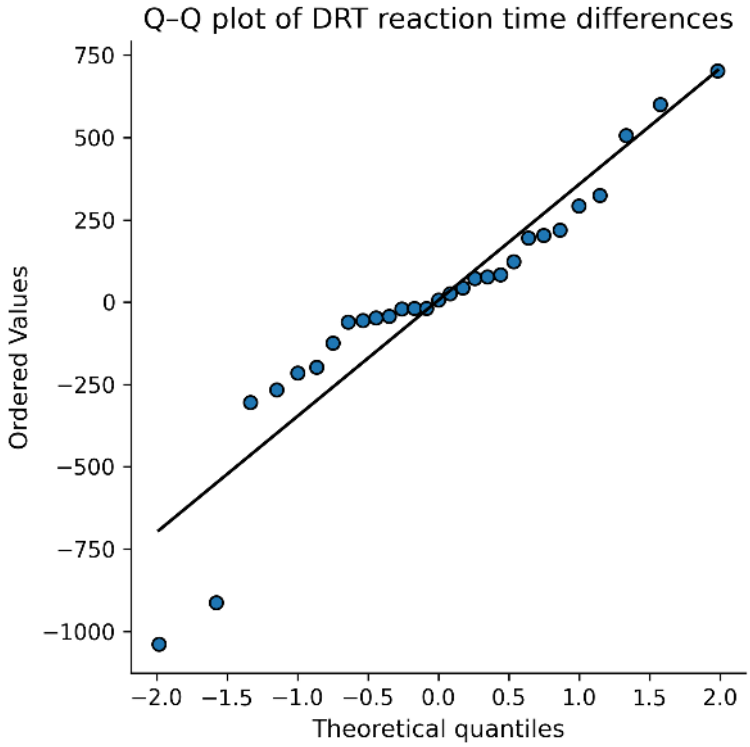

Figure 28: Q–Q plot of within-participant detection response task reaction time differences.

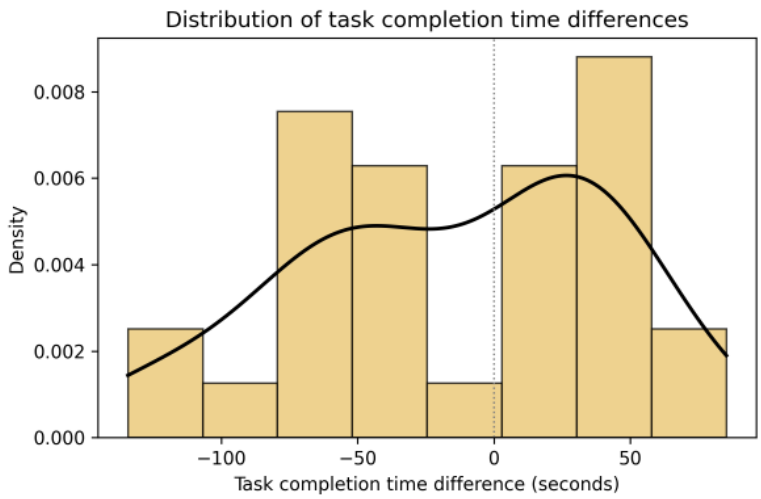

Figure 29: Histogram of within-participant task completion time differences.

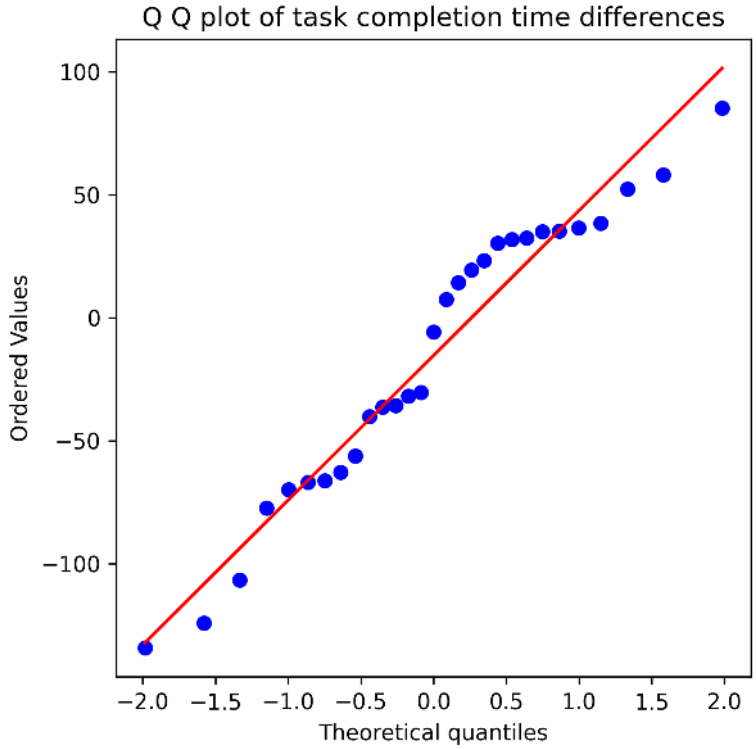

Figure 30: Q–Q plot of within-participant task completion time differences.

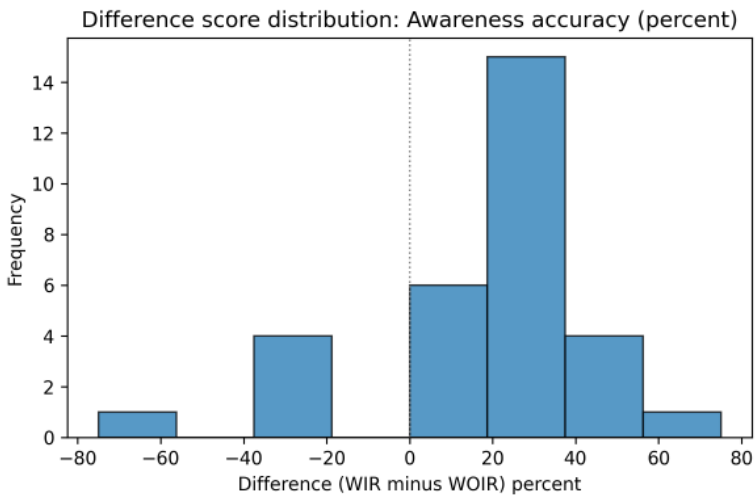

Figure 31: Histogram of within-participant SPAM probe accuracy differences.

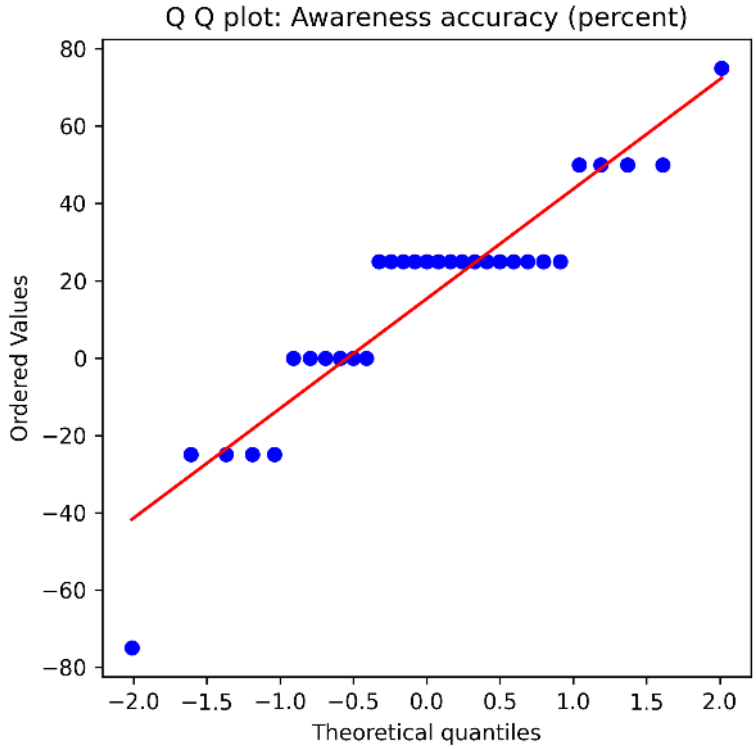

Figure 32: Q–Q plot of within-participant SPAM probe accuracy differences.

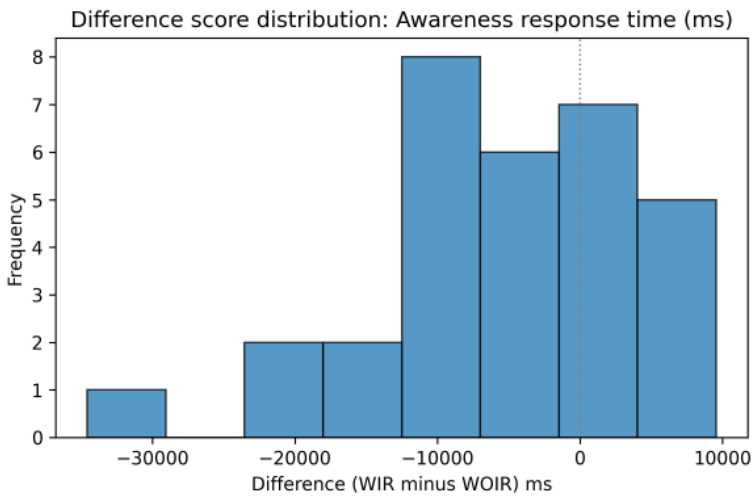

Figure 33: Histogram of within-participant SPAM probe response time differences.

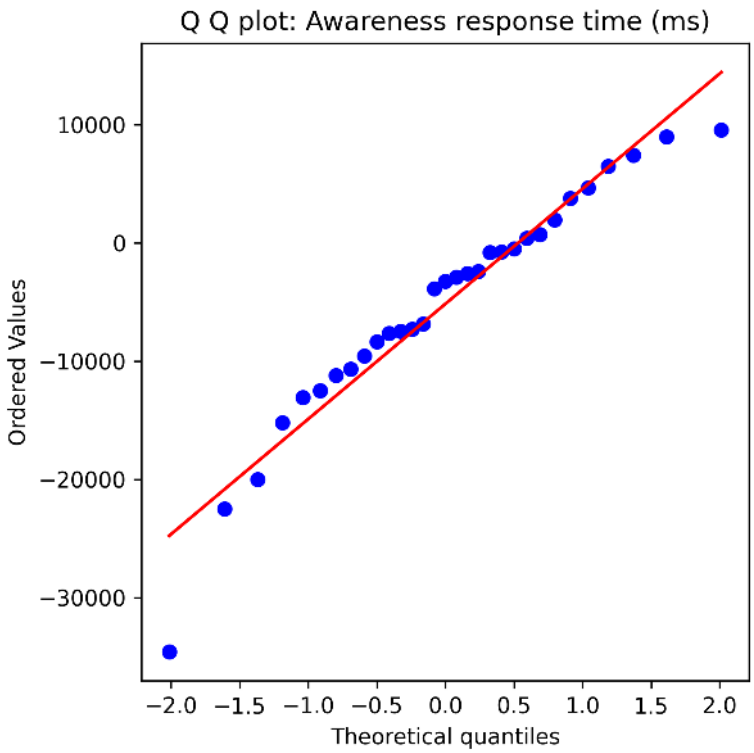

Figure 34: Q–Q plot of within-participant SPAM probe response time differences.

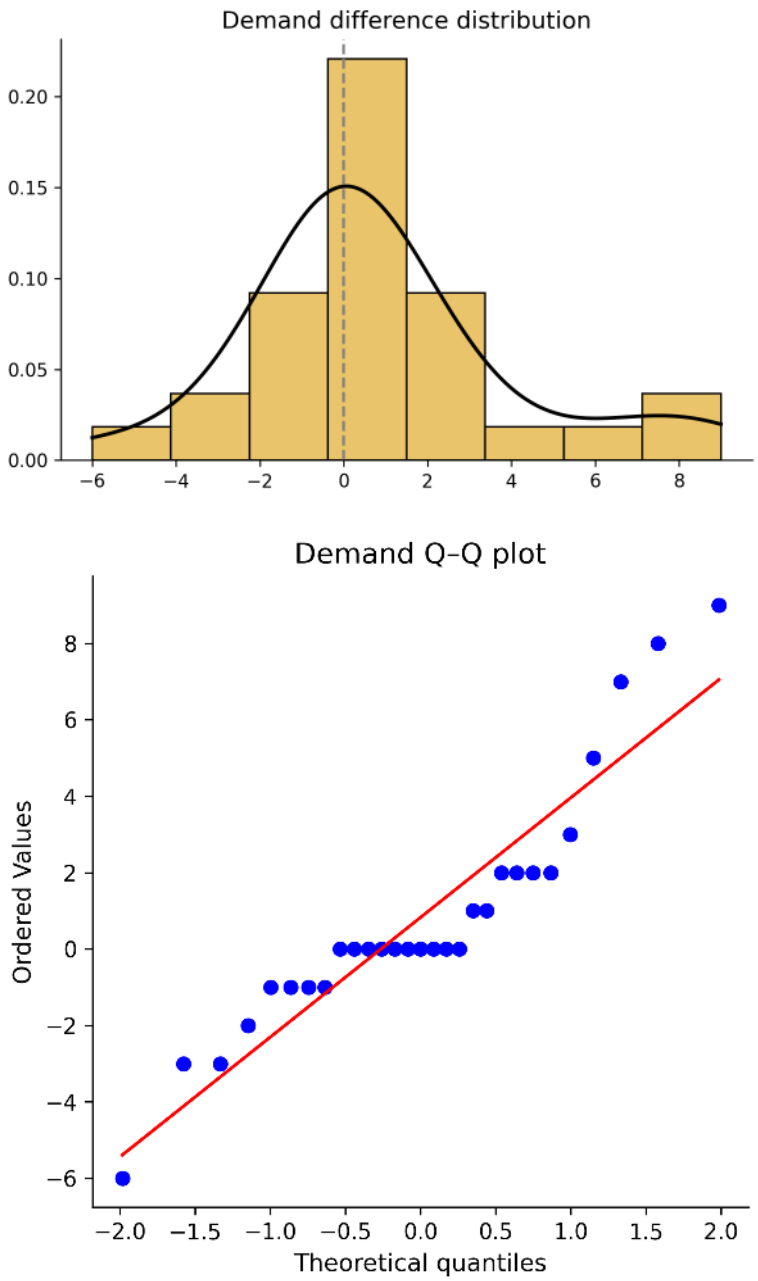

Figure 35: Histogram and Q–Q plot of paired difference scores for SART Demand.

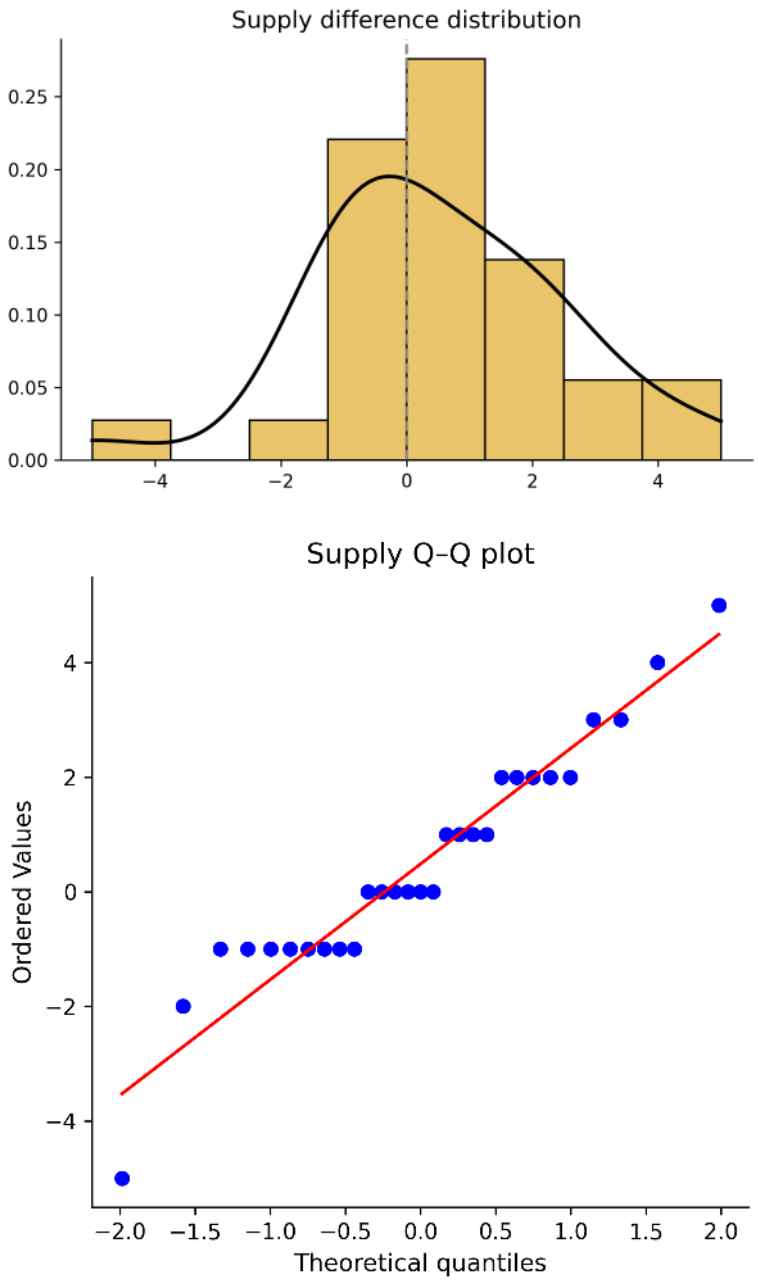

Figure 36: Histogram and Q–Q plot of paired difference scores for SART Supply.

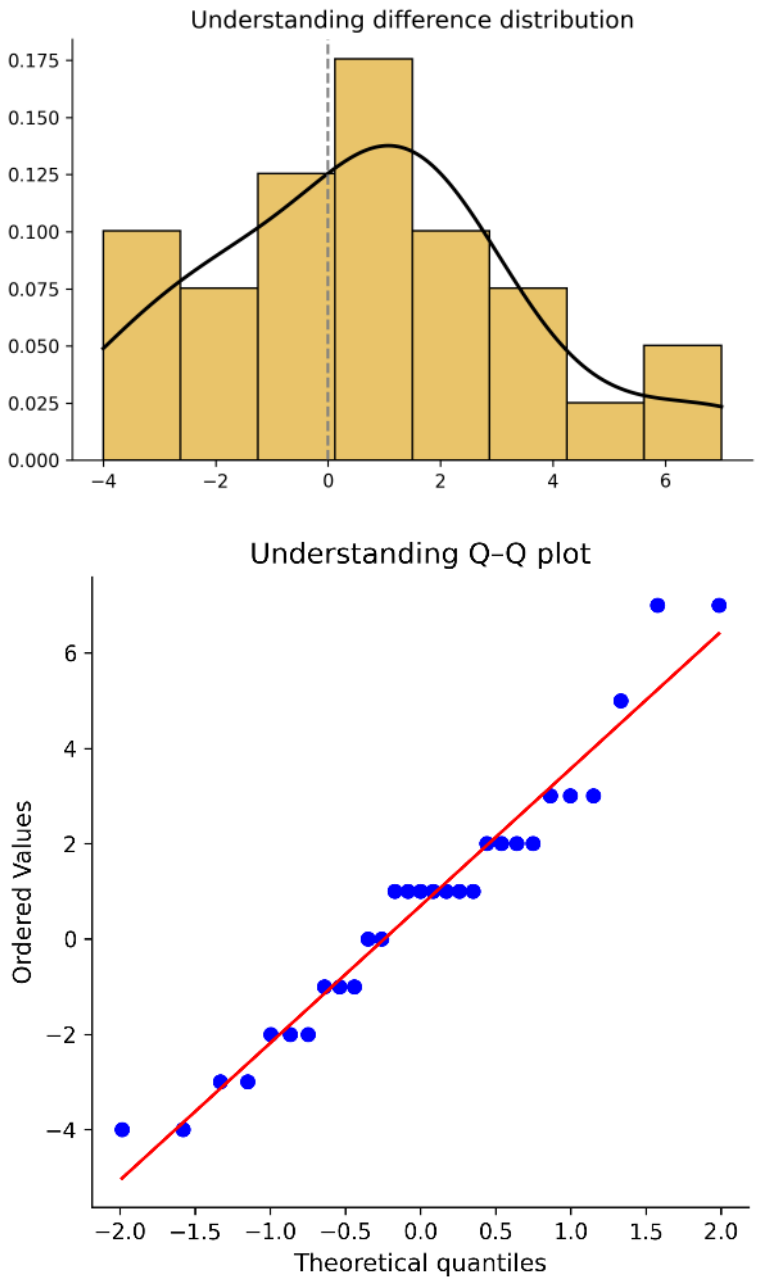

Figure 37: Histogram and Q–Q plot of paired difference scores for SART Understanding.

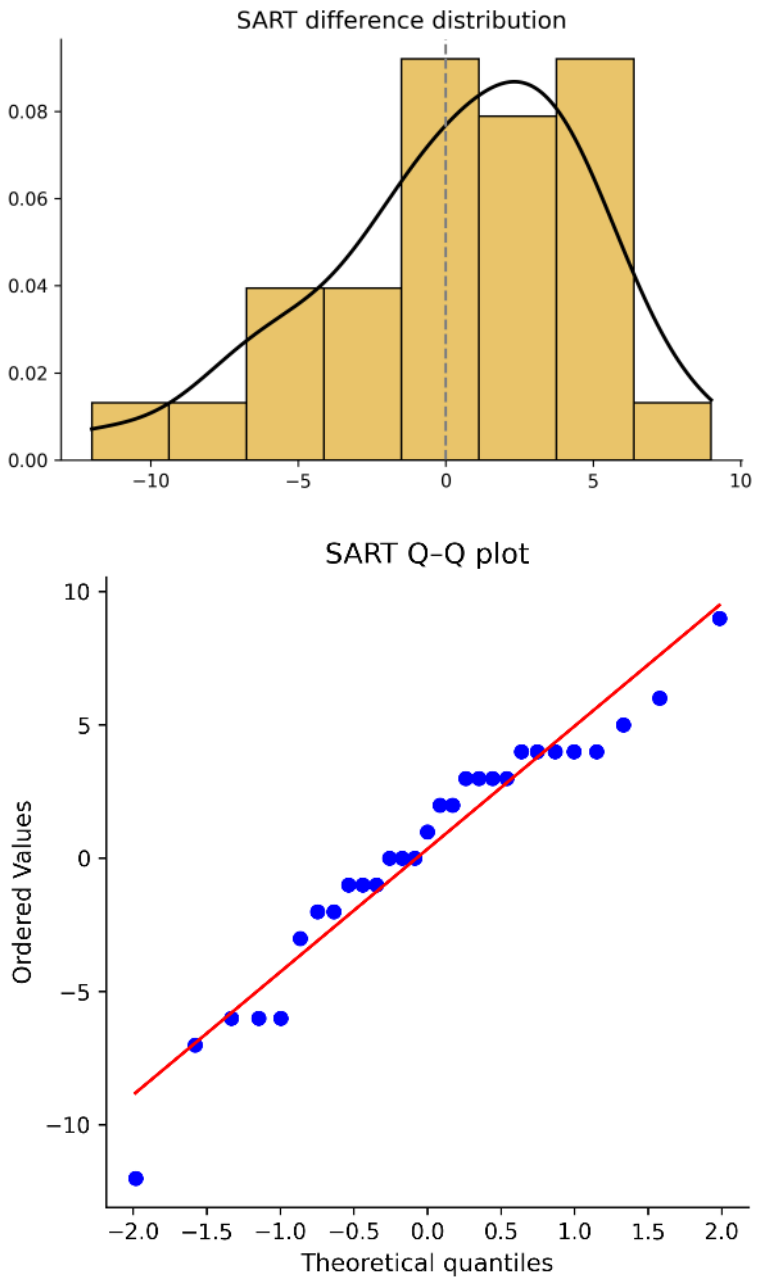

Figure 38: Histogram and Q–Q plot of paired difference scores for overall SART.

A.6. Participant Level SUS Scores

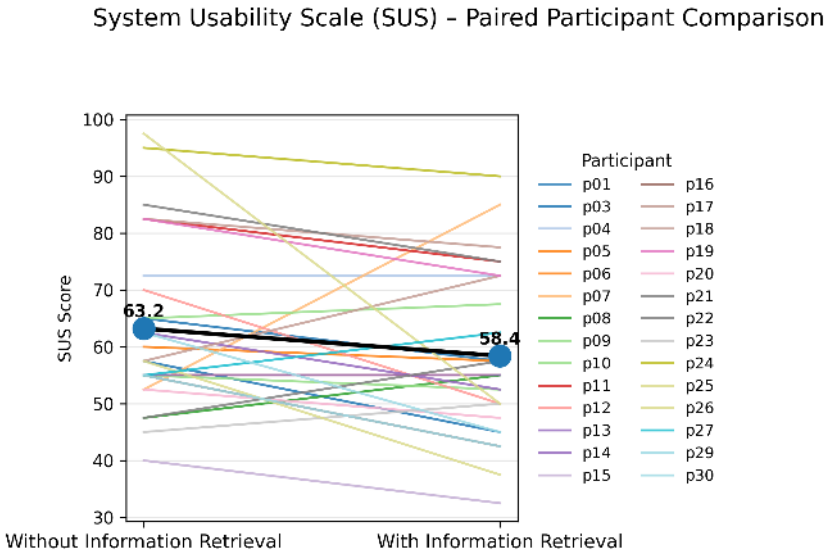

Figure 39: Participant-level paired System Usability Scale (SUS) scores.

A.7. Qualitative Code System Visualizations

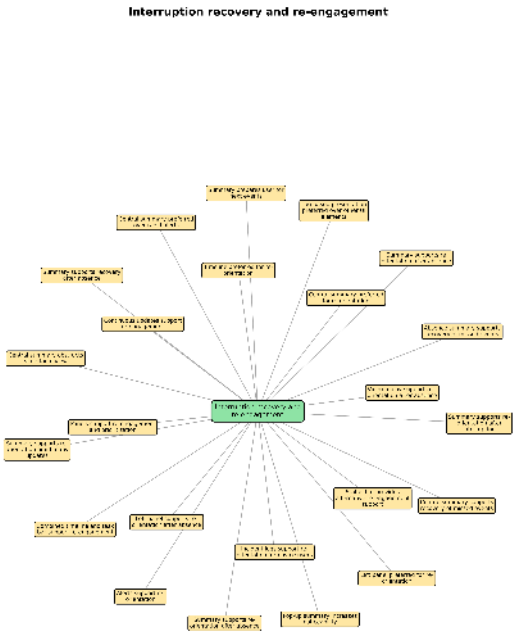

Figure 40: Sub codes associated with Theme 1: Structured information supports re-engagement after interruptions.

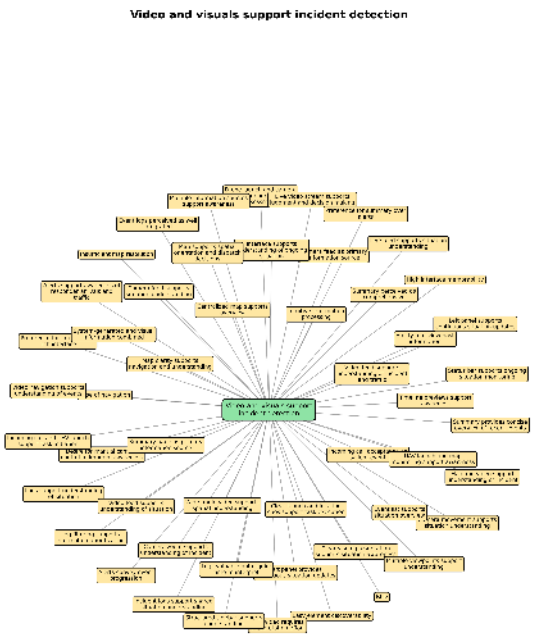

Figure 41: Sub codes associated with Theme 2: Video and visual elements supporting incident understanding.

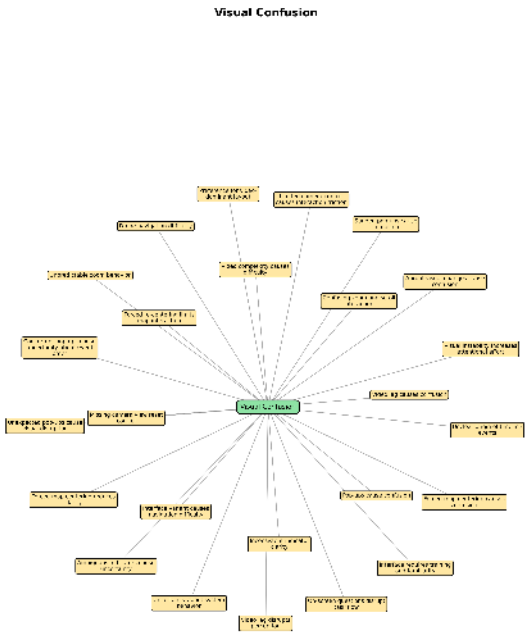

Figure 42: Sub codes associated with Theme 3: Visual complexity and confusion.

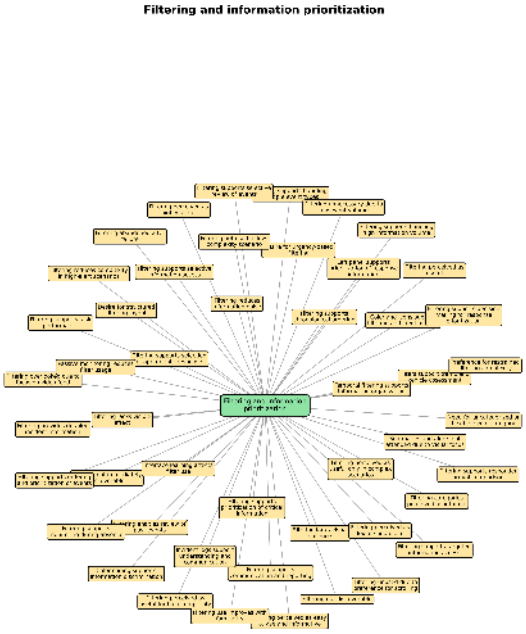

Figure 43: Sub codes associated with Theme 4: Filtering and information prioritization.

A.8. Qualitative Analysis Procedure

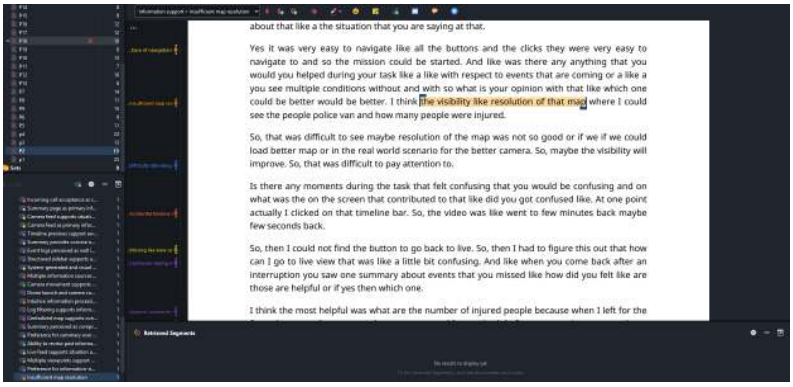

Figure 44: Example of transcript coding in MAXQDA showing initial codes applied to interview excerpts.

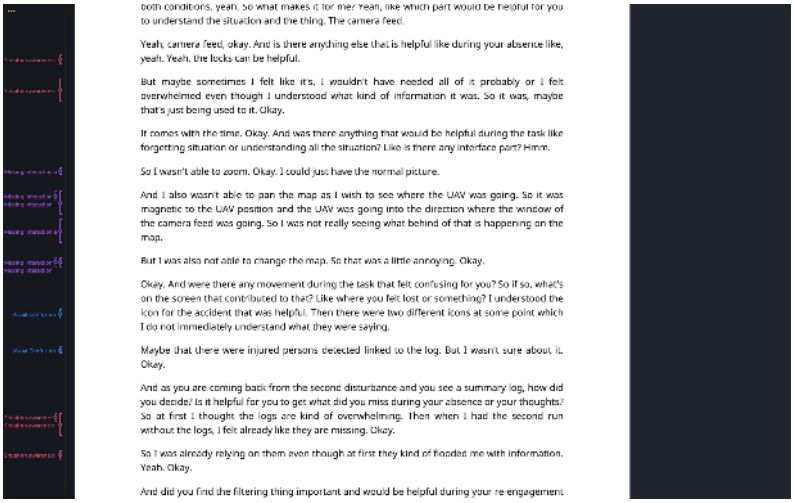

Figure 45: Further example of transcript coding in MAXQDA.

A.9. Analysis Code Repository

All quantitative analyses were conducted using custom Python scripts developed for this study. For transparency and reproducibility, the analysis codebase is available at:

https://github.com/HamzaMohsin1996/data-analysis-uav-interface

A.10. Interaction log data

This appendix documents the interaction log data collected during the experimental study.

All participant interactions recorded during the experimental trials are aggregated in a single file named `summary.csv`. The file contains interaction logs from all 29 participants and both experimental conditions (WIR and WOIR). Each row in the file corresponds to a single interaction or system event recorded during task execution.

The interaction logs include events related to emergency call handling, UAV mission initiation, drone monitoring, secondary task performance during interruptions, re engagement, workload probes, and situational awareness probes.

The file is provided as supplementary material accompanying this thesis and is not reproduced in full due to its size.

A.10.1. Log file structure

The complete interaction log file `summary.csv` is available for download via an online repository. The file can be accessed at the following link: https://drive.google.com/file/d/17lOo4JJHvIbF7KHzw-A5SsClqVcgYV3r/view?usp=sharing

- `file`: source log file corresponding to a single participant and condition
- `participant`: anonymized participant identifier
- `condition`: experimental condition (WIR or WOIR)
- `phase`: experimental phase during which the event occurred (e.g., task, interruption)
- `time`: timestamp in seconds relative to the start of the trial
- `topic`: type of interaction or system event (e.g., call events, DRT probes, secondary tasks, awareness probes)
- `data`: event specific metadata encoded as a structured object

A.11. Source code repositories

This appendix provides access to the source code repositories used to implement the experimental system described in this thesis. The code is provided for transparency and reproducibility and is not required to understand the experimental design or results.

A.11.1. Frontend interface implementation

The frontend interface used in the experimental study is implemented as a web based application. The source code corresponding to the frontend interface is available in the following public repository:

https://github.com/TTZ-Unbemannte-Flugsysteme/uav-info-retrieval-interface.git

The repository contains the frontend implementations corresponding to the experimental conditions described in this thesis. It includes code for interface layout, interaction handling, visualization of incident information, and communication with the simulation environment.

A.11.2. Simulation environment implementation

The simulation environment used to generate incident dynamics and visual content during the experiment is implemented using Unity. The source code corresponding to the simulation environment is available in the following public repository:

https://github.com/TTZ-Unbemannte-Flugsysteme/uav-info-retrieval-unity

The repository contains the Unity scenes and simulation logic used to generate incident events, drone behavior, and video streams presented to participants during the experimental trials.

A.12. Onboarding interface

This appendix provides illustrative examples of the digital onboarding interface used prior to the experimental tasks.

The onboarding was designed to familiarize participants with the simulated emergency context, the role of the UAV, and the core interface elements before data collection began. The onboarding content was identical for all participants and conditions. No data from the onboarding phase were included in the analysis.

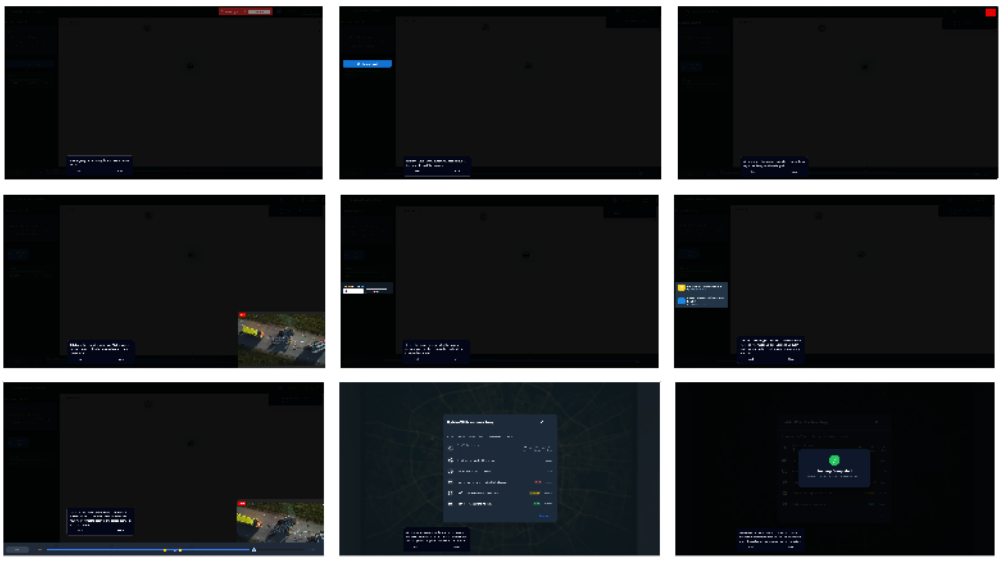

Figure 46: Digital onboarding phase introducing the simulated emergency scenario and core interface elements.

A.13. Online Questionnaire and Response Data

All participant data were collected using an online questionnaire implemented with Google Forms. The questionnaire was used to collect demographic information and background characteristics.

Anonymized response data (student participants):
https://docs.google.com/spreadsheets/d/1_bkFnSM4oFeGxsqq-GMOeHAoa8WePJm_rDdBnOD8tz4

Anonymized response data (professional dispatchers, $n = 2$):
https://docs.google.com/spreadsheets/d/1KF5hEjMrIrn_9K6AOBBazczaUMy2Q8u7Wj471LGShyI

A.14. Design Artifacts and Prototypes

This appendix provides access to external design artifacts created during the design process of the dispatcher interface. These artifacts are provided for transparency and documentation purposes and were not used directly during the experimental trials.

A.14.1. Figma design boards

Early design exploration, information structuring, and interface concept development were conducted using Figma. The complete design board used during the information gathering and conceptual design phase is available at the following link:

https://www.figma.com/board/PqlGVEKGKg0PVcB7fMdemd/Information-Gathering?node-id=5699-745&t=L5xE5sldSQ2DR23q-1

A.14.2. Interface prototype videos

Screen recordings of interactive interface prototypes were created to document design iterations and interaction concepts prior to implementation. The videos illustrate navigation, information presentation, and re-engagement mechanisms explored during the design phase.

The prototype videos are available in the following shared folder:

https://drive.google.com/drive/folders/1cjECngR4XOlKAiH--811FcNn5G_fPKta?usp=sharing